

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2, Class 2

ASME Code Version : 2023

Analysis Performed by : SL User

Job File : ...\\401-E-31004-AK.pvdb

Date of Analysis : May 21, 2026 5:03pm

PV Elite 27, January 2025

Vessel Design Summary:

ASME Code, Section VIII Division 2 Class 2, 2023

Diameter Spec : 1100.000 mm ID
 Vessel Design Length, Tangent to Tangent 573.87 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 270 °C
 Channel Side Design Temperature 300 °C
 Shell Side Design Pressure 57.105 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 67.749 kgf/cm²
 Shop Channel Side Test Pressure 197.436 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA-387 11	2	...	K11789	No	Yes
Shell	SA-516 70	...	...	K02700	No	No
Head	SA-516 70	...	...	K02700	No	No
Flange	SA-336 F11	2	...	K11572	No	Yes
Flange	SA-266 2	...	...	K03506	No	No
Nozzle	SA-336 F11	2	...	K11572	No	No
Nozzle	SA-106 B	...	...	K03006	No	No
Re-Pad	SA-516 70	...	...	K02700	No	No
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-213 T11	...	...	K11597	No	Yes
Tubesheet	SA-336 F11	2	...	K11572	No	Yes
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	No	No
Hrz Bolting	SA-193 B7	...	<= 64	G41400	No	No
Saddles	SA-516 70	...	...	K02700	No	No

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch Cover	163.322	0.00	173.000	1.5000	N/A	No
Ch Flg	163.268	0.00	183.000	1.5000	N/A	No
LHS Ch Shell	163.269	0.00	185.000	1.5000	N/A	No
LHS Shell Flg	57.215	1.05	59.800	1.5000	N/A	No
Main Shell	57.216	1.05	59.800	1.5000	N/A	No
Dish End	57.216	1.05	70.000	1.5000	No	No

Liquid Level: 164.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	30.1	29.5	303.0	295.0	286.8	199.3	1.00	1.00
Body Flg	72.5	42.0	328.0	320.0	312.6	312.6	1.00	1.00
Cylinder	222.5	150.0	66.0	66.0	58.1	...	1.00	1.00
Body Flg	268.8	22.7	183.0	175.0	154.2	154.2	1.00	1.00
Cylinder	568.8	300.0	25.0	25.0	23.9	6.9	1.00	1.00



Ellipse	573.9	5.1	32.0	26.0	21.5	4.2	1.00	1.00
---------	-------	-----	------	------	------	-----	------	------

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.954	Shell	Passed
N2-Tube Outlet	WRC-297	0.954	Shell	Passed
N3-Shell Inlet	WRC-297	0.955	Shell	Passed
N4-Shell outle	WRC-297	0.954	Shell	Passed

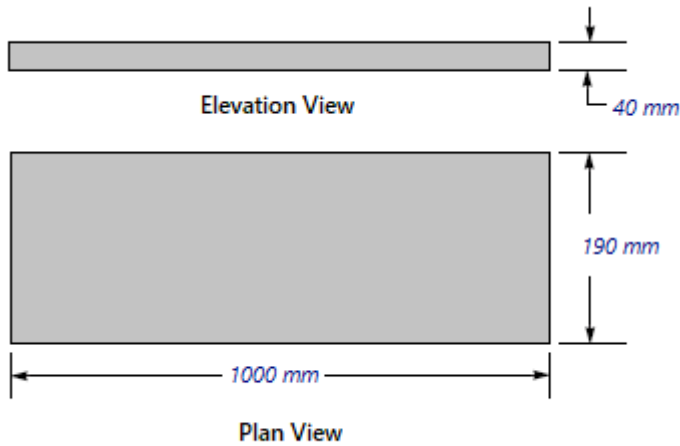
Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1000.000 mm
Wear Pad Width	380.000 mm
Wear Pad Thickness	16.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	750.000 mm

Baseplate Length	1000.000 mm
Baseplate Thickness	40.000 mm
Baseplate Width	190.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	35.000 mm
Web Thickness	30.000 mm
Height of Center Web	260.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	57868.88 kgf
Maximum Transverse Saddle Shear Load	1434.53 kgf
Maximum Longitudinal Saddle Shear Load	2869.07 kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	22110.87 kgf
Maximum Transverse Saddle Shear Load	287.03 kgf
Maximum Longitudinal Saddle Shear Load	152.52 kgf

Weights:

Fabricated - Bare w/o removable internals	26444.4 kg
Shop Test - Fabricated + water (full)	31116.0 kg
Shipping - Fab. + rem. intls.+ shipping app.	26962.5 kg



Erected	- Fab. + rem. intls.+ insul. (etc)	26962.5	kg
Empty	- Fab. + intls. + details + wghts.	26962.5	kg
Operating	- Empty + operating liquid (no ca)	31634.0	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N1-Tube Inlet	172	...	...	OK	OK	Passed
N2-Tube Outlet	172	...	...	OK	OK	Passed
N3-Shell Inlet	59.4	OK	...	OK	OK	Passed
N4-Shell outlet	59.5	OK	...	OK	OK	Passed
K1-Vent	58.9	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 58.9 Nozzle : K1-Vent [Shellside]
 Minimum MAWP Nozzles : 172 Nozzle : N2-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N1-Tube Inlet	1474.525	90.000	255.500
30	N2-Tube Outlet	1474.525	270.000	255.500
50	N3-Shell Inlet	2987.875	270.000	74.665
50	N4-Shell outlet	2987.875	90.000	74.665
50	K1-Vent	5187.875	90.000	21.972

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1-Vent	50 mm	XXS	LWN	60.3	11.074	...	...	325	600
N3-Shell Inlet	150 mm	80	WNF	168.3	10.973	268.27	26.00	331	600
N4-Shell outlet	150 mm	80	WNF	168.3	10.973	268.27	26.00	331	600
N1-Tube Inlet	508 mm	Actual	None	584.2	38.100	...	...	542	...
N2-Tube Outlet	508 mm	Actual	None	584.2	38.100	...	...	542	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-Vent	SA-106 B	11.074	16.000	...	...	...
N3-Shell In	SA-106 B	25.000	10.000	20.000	26.000	...
N4-Shell ou	SA-106 B	25.000	10.000	20.000	26.000	...
N1-Tube Inl	SA-336 F11	66.000	10.000	...	...	...
N2-Tube Out	SA-336 F11	66.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1-Vent	518.788	90.0	300.00	0.00	Main Shell
N3-Shell Inlet	298.788	270.0	300.00	0.00	Main Shell
N4-Shell outlet	298.788	90.0	300.00	0.00	Main Shell
N1-Tube Inlet	147.452	90.0	300.00	0.00	LHS Ch Shell
N2-Tube Outlet	147.452	270.0	300.00	0.00	LHS Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 303.0mm THK X 1680.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 328.0mm THK X 1680.0mm OD	SA-336 F11
1	CYLINDER: 66.0mm THK X 1100.0mm ID X 150.0cm	SA-387 11
1	WELD NECK FLANGE: 183.0mm THK X 1430.0mm OD	SA-266 2
1	CYLINDER: 25.0mm THK X 1100.0mm ID X 300.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm THK X 1100.0mm ID X 5.1cm	SA-516 70
1	INSULATION: 290mm X 105mm THK	Hot Insulation
1	INSULATION: 410mm X 105mm THK	Hot Insulation
1	INSULATION: 1500mm X 105mm THK	Hot Insulation
5	BAFFLES: 1092.0mm DIA X 10.0mm THK	SA-516 70
8	TIE RODS: 14mm DIA X 1749mm	SA-193 B7M
1	TUBESHEET: 1430.0mm X 230.0mm THK	SA-336 F11
1452	TUBES: 2600.0mm X 19.0mm DIA X 2.5mm THK	SA-213 T11
1	INSULATION: 227mm X 145mm THK	Hot Insulation
2	SADDLE: 150.0mm X 120 DEG	SA-516 70
1	INSULATION: 3000mm X 145mm THK	Hot Insulation
1	INSULATION: 325mm X 145mm THK	Hot Insulation
2	GASKET: 1164.0mm OD X 1132.0mm ID	...
24	BODY FLANGE BOLTS: 95.2mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 95.2mm DIA	...
36	BODY FLANGE BOLTS: 48.0mm DIA	SA-193 B7
36	NUTS FOR BODY FLANGE BOLTS: 48.0mm DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
LHS Shell Flg[11]	B	14	3	No	0.743	25.000	1.00
Bolting [21]		-48					
Main Shell [8]	B	14	5	No	0.788	25.000	1.00
Dish End [10]	B	14	1	No	0.674	26.000	1.00
Dish End [7]	B	19	3	No	0.607	32.000	1.00
Left Saddle [24]	B	6	6	No	1.000	16.000	1.00
New Saddle [24]	B	6	6	No	1.000	16.000	1.00
N3-Shell Inlet[1]	B	14	4	No	0.765	25.000	1.00
Nozzle Flg [5]	A	-18	-45	No			
N4-Shell outle[1]	B	14	4	No	0.763	25.000	1.00
Nozzle Flg [5]	A	-18	-45	No			
K1-Vent [1]	B	-104	-104	No	0.130	9.690	1.00
Nozzle Flg [5]	A	-18	-45	No			
Nozl. Bolting[22]		-48		No	1.000	64.000	1.00
Tubesheet: SS[13]	!	11		No			
Warmest MDMT:		19	6				
Ch Cover [11]	!	11		Yes			
Ch Flg [11]	B	14	-1	Yes	0.735	66.000	1.00
LHS Ch Shell [8]	!	11		Yes			
N1-Tube Inlet [1]	B	14	-4	Yes	0.687	66.000	1.00
N2-Tube Outlet[1]	B	14	-4	Yes	0.688	66.000	1.00
Tubesheet: CS[14]	!	11		Yes			
Bolting [21]		-40					
Warmest MDMT:		14	-1				
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail		
Shell	11.0		11.4		Pass		
Channel/Tube	11.0		11.4		Pass		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	57.105	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	270.0	°C
Channel Side Design Temperature	300.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	User Defined Hydro
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------



0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g_0 is taken as $\min(g_0, \text{attached shell thk})$ Yes

Using Metric Material Databases, ASME II D Yes

Using ASME VIII-1 Appendix 46 No

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Ch Cover
Distance "FROM" to "TO"	29.5 cm
Flange Outside Diameter	1680 mm
Element Thickness	295 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	303 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	300 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11 [Impact Tested]
Allowable Stress, Ambient	1876.2 kgf/cm ²
Allowable Stress, Operating	1488.7 kgf/cm ²
Allowable Stress, Hydrotest	2663.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2233.1 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-2



UNS Number K11572
 Class / Thickness / Grade 2
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 10
 Detail Type Liquid
 Detail ID Liquid: 10
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 164 cm
 Liquid Density 999.54 kg/m³

Element From Node 10
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 29 cm
 Thickness of Insulation 105 mm
 Density 0.00015 kg/cm³

Element# 2/6
 Element From Node 20
 Element To Node 30
 Element Type Flange
 Description Ch Flg
 Distance "FROM" to "TO" 42 cm
 Flange Inside Diameter 1100 mm
 Element Thickness 320 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 328 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 163.16 kgf/cm²
 Design Temperature Internal Pressure 300 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 300 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-336 F11
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 110 cm
 Liquid Density 999.54 kg/m³

Element From Node 20
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 41 cm
 Thickness of Insulation 105 mm
 Density 0.00015 kg/cm³

Element# 3/6
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description LHS Ch Shell
 Distance "FROM" to "TO" 150 cm
 Inside Diameter 1100 mm
 Element Thickness 66 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 66 mm



External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	300	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	[Impact Tested]
Allowable Stress, Ambient	2110.7	kgf/cm ²
Allowable Stress, Operating	1672.3	kgf/cm ²
Allowable Stress, Hydrotest	3002.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2508.4	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	110 cm
Liquid Density	999.54 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	150 cm
Thickness of Insulation	105 mm
Density	0.00015 kg/cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	75 cm
Nozzle Diameter	508 mm
Nozzle Schedule	120
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	578.48 kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element From Node	30
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	75 cm
Nozzle Diameter	508 mm
Nozzle Schedule	120
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	641.28 kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange
Description	LHS Shell Flg



Distance "FROM" to "TO"	22.7	cm
Flange Inside Diameter	1100	mm
Element Thickness	175	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	183	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	57.105	kgf/cm ²
Design Temperature Internal Pressure	270	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	270	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	1682.4	kgf/cm ²
Allowable Stress, Operating	1362.3	kgf/cm ²
Allowable Stress, Hydrotest	2421.7	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2039.4	kgf/cm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	110	cm
Liquid Density	999.54	kg/m ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	22.7	cm
Thickness of Insulation	145	mm
Density	0.00015	kg/cm ³

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	300 cm
Inside Diameter	1100 mm
Element Thickness	25 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	25 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	57.105 kgf/cm ²
Design Temperature Internal Pressure	270 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1784.4 kgf/cm ²
Allowable Stress, Operating	1435.7 kgf/cm ²
Allowable Stress, Hydrotest	2518.6 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2153.6 kgf/cm ²
MDMT Curve	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0



Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 75 cm
 Width of Saddle 150 mm
 Height of Saddle at Bottom 1000 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 380 mm
 Thickness of Wear Plate 16 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.30000001
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 50
 Detail Type Saddle
 Detail ID New Saddle
 Dist. from "FROM" Node / Offset dist 265 cm
 Width of Saddle 150 mm
 Height of Saddle at Bottom 1000 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 380 mm
 Thickness of Wear Plate 16 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 110 cm
 Liquid Density 999.54 kg/m³

Element From Node 50
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 300 cm
 Thickness of Insulation 145 mm
 Density 0.00015 kg/cm³

Element From Node 50
 Detail Type Nozzle
 Detail ID N3-Shell Inlet
 Dist. from "FROM" Node / Offset dist 30 cm
 Nozzle Diameter 150 mm
 Nozzle Schedule 80
 Nozzle Class 600
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 57.695 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 50
 Detail Type Nozzle
 Detail ID N4-Shell outlet



Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	150	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	57.695	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K1-Vent	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50	mm
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.6213	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Dish End	
Distance "FROM" to "TO"	5.08	cm
Inside Diameter	1100	mm
Element Thickness	26	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	57.105	kgf/cm ²
Design Temperature Internal Pressure	270	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	270	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	110	cm
Liquid Density	999.54	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	32.58	cm
Thickness of Insulation	145	mm
Density	0.00015	kg/cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Ch Cover		30.135	...	29.5	...
Ch Flg		72.4525	...	42	...
LHS Ch Shell		222.452	...	150	...
LHS Shell Flg		268.787	...	22.7	...
Main Shell		568.788	...	300	...
Dish End		573.867	...	5.08	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

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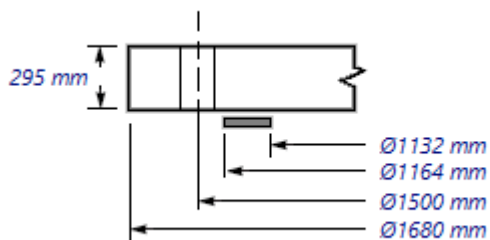


Flange Input Data Values

Description: New Flange :

Ch Cover

Flange Type		Blind	
Design Pressure	P	163.32	kgf/cm ²
Design Temperature		300	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1680.000	mm
Flange Thickness	t	295.0000	mm
Flange Material		SA-336 F11	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1488.73	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Perimeter along the Center of the Bolts	L	4712.389	mm
Diameter of Bolt Circle	C	1500.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1175.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	1164.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm
Flange Face Nubbin Width	w	15.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1149.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$\begin{aligned}
 &= \pi/4 \cdot G^2 \cdot Peq \\
 &= \pi/4 \cdot 1150^2 \cdot 163 \\
 &= 1695657.375 \text{ kgf}
 \end{aligned}$$



Contact Load on Gasket Surfaces [Hp]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.13 \cdot \pi \cdot 1150 \cdot 3 \cdot 163 \\ &= 252278.062 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(H + Hp + H'p, 0) \\ &= \max(1695657 + 252278 + 0, 0) \\ &= 1947935.375 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 703 \cdot 7.13 \cdot \pi \cdot 1150 + 0 \cdot 0 \cdot 0 \\ &= 181001.078 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa) \\ &= \max((1947935 + 0 + 4 \cdot 0/1150)/1621, 181001/1621) \\ &= 1201.494 \text{ cm}^2 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1201.494	1565.107	
Radial Distance between Bolts and Edge:	88.900	90.000	
Circ. Spacing between the Bolts:	193.675	195.789	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= Ab \cdot Sa / (y \cdot \pi (Go + Gi)) \\ &= 1565 \cdot 1621 / (703 \cdot \pi (1164 + 1132)) \\ &= 50.035 \text{ mm [Note: Exceeds actual gasket width, 16.000]} \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$\begin{aligned} &= Sa \cdot Ab \\ &= 1621 \cdot 1565 \\ &= 2537446.25 \text{ kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= Wo \\ &= 1947935.38 \text{ kgf} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (1500 - 1150) / 2 \\ &= 175.1274 \text{ mm} \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned} &= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\ &= 1.9 \cdot 1947935 \cdot 175 \cdot 1 / (293^2 \cdot 1150) + 0.3 \cdot 1 \cdot 163 \cdot 1150^2 / 293^2 \\ &= 1406.32 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned} &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\ &= 1.9 \cdot 2537446 \cdot 175 \cdot 1 / (293^2 \cdot 1150) \\ &= 852.49 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Operating [BSO]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 1947935 / 1565 \\ &= 1244.60 \text{ kgf/cm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (181001 / 1565) \\ &= 115.65 \text{ kgf/cm}^2 \end{aligned}$$



Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1406.32	1488.73	852.49 1876.18
Bolting	1244.60	1621.26	115.65 1621.26

Reqd. Blind Flange Thickness at Center 286.761 mm
 Reqd. Blind Flange Thickness at Gasket 199.340 mm

Estimated M.A.W.P. (Operating) 173 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 5068.2 kg
 Estimated Unfinished Weight of Forging at given Thk 5068.2 kg

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1392.06	1876.18	843.84 1876.18
Bolting	1244.60	1621.26	115.65 1621.26

Minimum Required Flange Thickness, no Corrosion: 254.127 mm

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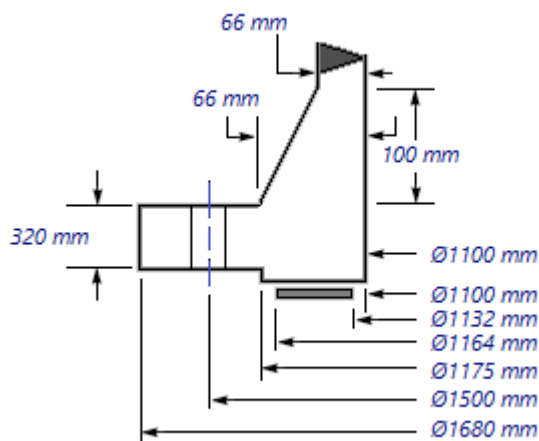


Flange Input Data Values

Description: Ch Flg :

Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	163.27	kgf/cm ²
Design Temperature		300	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1680.000	mm
Flange Thickness	t	320.0000	mm
Thickness of Hub at Small End	go	66.0000	mm
Thickness of Hub at Large End	gl	66.0000	mm
Length of Hub	h	100.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	1488.73	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Diameter of Bolt Circle	C	1500.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1175.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1164.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	3.1750	mm
Flange Face Nubbin Width	w	15.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$



$$= (163(1100+1.5)/(1672-0.5 \cdot 163)) + (1.5)$$

$$= 58.1079 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1672 \cdot \log((2 \cdot 64.5 + 552) / 552)$$

$$= 184.962 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 2111 \cdot \log((2.0 \cdot 66 + 550) / 550)$$

$$= 239.203 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	318.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1103.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	64.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	64.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	134.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1149.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 1150^2 \cdot 163$$

$$= 1695097.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1150 \cdot 3 \cdot 163$$

$$= 252194.688 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1103^2 \cdot 163$$

$$= 1560064.000 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1695097 - 1560064$$

$$= 135032.984 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(1695097 + 252195 + 0, 0)$$

$$= 1947291.750 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 7.13 \cdot \pi \cdot 1150 + 0 \cdot 0 \cdot 0$$

$$= 181001.078 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1947291 + 0 + 4 \cdot 0 / 1150) / 1621, 181001 / 1621)$$

$$= 1201.097 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 95.2 + 6 \cdot 318 / (3 + 0.5)$$

$$= 736.500 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1500 \cdot \sin(\pi / 24)$$

$$= 195.789 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{196/(2 \cdot 95.2 + 318)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1201.097	1565.107	
Radial Distance between Hub and Bolts:	112.713	134.000	
Radial Distance between Bolts and Edge:	88.900	90.000	
Circ. Spacing between the Bolts:	193.675	195.789	736.500

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1565 \cdot 1621 / (703 \cdot \pi(1164 + 1132))$$

$$= 50.035 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1565$$

$$= 2537446.25 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1947291 - 1695097$$

$$= 252194.77 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1500 - 1150) / 2$$

$$= 175.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (134 + 64.5 + 175) / 2$$

$$= 186.8137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 134 + (64.5 / 2)$$

$$= 166.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1560064.	166.2500	1.0000	259361.
Face Pressure, Mt	135033.	186.8137	1.0000	25226.
Gasket Load, Mg	252195.	175.1274	1.0000	44166.
Gasket Seating, Matm	2537446.	175.1274	1.0000	444377.

Total Moment for Operation, Mop 328753. kgf-m.
 Total Moment for Gasket seating, Matm 444377. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 266.727 mm
 Hub Ratio, h/h0 = Defined as 2 2.000
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.000

Note: Because g1 = g0, the ratio h/h0 has been set to 2.



Flange Factors for Integral Flange:

Factor F per 4.16.5 0.926
 Factor V per 4.16.5 0.549
 Factor f per 4.16.5 1.000

Factors from Table 4-16.4

T = 1.701 K = 1.523
 U = 5.261
 Y = 4.788 Z = 2.515
 e = 0.0035 1/mm d = 0.10632E+08 mm³

Stress Factors

Beta = 2.474 Alpha = 2.106
 Gamma = 1.238
 Delta = 3.039 Lambda = 4.277

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g¹²)
 = (1•328753/1103)/(4.28•64.5²)
 = 1675.06 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g¹²)
 = (1•444377/1103)/(4.28•64.5²)
 = 2264.18 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= (β • Mop / Bcor) / (L • t²)
 = (2.47•328753/1103)/(4.28•318²)
 = 169.96 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= (β • Matm/Bcor) / (L • t²)
 = (2.47•444377/1103)/(4.28•318²)
 = 229.73 kgf/cm²

Tangential Flange Stress, Operating [STo]:

= (Y • Mo / (t² • Bcor)) - Z • SRo
 = (4.79•328753/(318²•1103))-2.52•170
 = 979.25 kgf/cm²

Tangential Flange Stress, Seating [STa]:

= (y • Matm / (t² • Bcor)) - Z • SRa
 = (4.79•444377/(318²•1103))-2.52•230
 = 1323.66 kgf/cm²

Average Flange Stress, Operating [SAo]:

= (SHo + max(SRo, STo)) / 2
 = (1675+max(170,979))/2
 = 1327.15 kgf/cm²

Average Flange Stress, Seating [SAa]:

= (SHa + max(SRa, STa)) / 2
 = (2264+max(230,1324))/2
 = 1793.92 kgf/cm²

Bolt Stress, Operating [BSo]:

= Wm1 / Ab
 = 1947291/1565
 = 1244.19 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (181001/1565)
 = 115.65 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1675.06	2233.09	2264.18	2814.27
Radial Flange	169.96	1488.73	229.73	1876.18



Tangential Flange	979.25	1488.73	1323.66	1876.18
Maximum Average	1327.15	1488.73	1793.92	1876.18
Bolting	1244.19	1621.26	115.65	1621.26

Minimum Required Flange Thickness 312.598 mm

Estimated M.A.W.P. (Operating) 183 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3328.1 kg
 Estimated Unfinished Weight of Forging at given Thk 4122.2 kg

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 444377 / 1 \cdot 100000 \cdot 0.55 / (4.05 \cdot 2081087 \cdot 64.5^2 \cdot 267 \cdot 0.3)$
 $= 0.453$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 328753 / 1 \cdot 100000 \cdot 0.55 / (4.05 \cdot 1901117 \cdot 64.5^2 \cdot 267 \cdot 0.3)$
 $= 0.366$ (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 444377 / 1 \cdot 100000 \cdot 0.55 / (4.28 \cdot 2081087 \cdot 64.5^2 \cdot 267 \cdot 0.3)$
 $= 0.429$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 328753 / 1 \cdot 100000 \cdot 0.55 / (4.28 \cdot 1901117 \cdot 64.5^2 \cdot 267 \cdot 0.3)$
 $= 0.347$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -1 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 15 °C
 The computed stress ratio is : 0.735

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1653.40	2814.27	2225.12	2814.27
Radial Flange	173.45	1876.18	233.42	1876.18
Tangential Flange	961.10	1876.18	1293.43	1876.18
Maximum Average	1307.25	1876.18	1759.28	1876.18
Bolting	1244.19	1621.26	115.65	1621.26

Minimum Required Flange Thickness, no Corrosion: 309.194 mm

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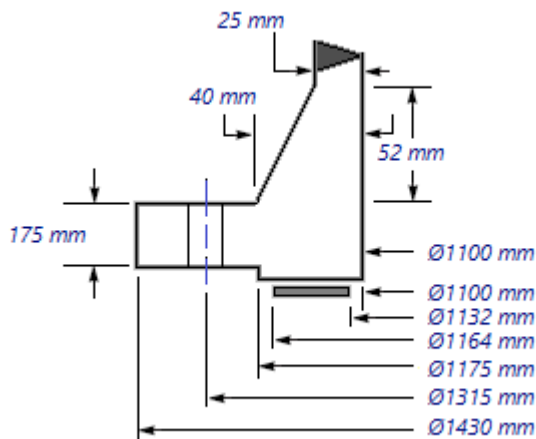


Flange Input Data Values

Description: Shell Flg :

LHS Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	57.22	kgf/cm ²
Design Temperature		270	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1430.000	mm
Flange Thickness	t	175.0000	mm
Thickness of Hub at Small End	go	25.0000	mm
Thickness of Hub at Large End	gl	40.0000	mm
Length of Hub	h	52.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1362.30	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	1315.000	mm
Nominal Bolt Diameter	a	48.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1175.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	1164.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm
Flange Face Nubbin Width	w	15.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$



$$= (57.2(1100+1.5)/(1436-0.5 \cdot 57.2)) + (1.5)$$
$$= 23.9248 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$
$$= 1436 \cdot \log((2 \cdot 23.5 + 552) / 552)$$
$$= 59.910 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$
$$= 1784 \cdot \log((2.0 \cdot 25 + 550) / 550)$$
$$= 79.320 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	173.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1103.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	38.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	23.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	67.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1149.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$
$$= \pi/4 \cdot 1150^2 \cdot 57.2$$
$$= 594025.875 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$
$$= 2 \cdot 7.13 \cdot \pi \cdot 1150 \cdot 0 \cdot 57.2$$
$$= 0.000 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$
$$= \pi/4 \cdot 1103^2 \cdot 57.2$$
$$= 546705.250 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$
$$= 594026 - 546705$$
$$= 47320.645 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(H + H_p + H'p, 0)$$
$$= \max(594026 + 0 + 0, 0)$$
$$= 594025.875 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$
$$= 0 \cdot 7.13 \cdot \pi \cdot 1150 + 0 \cdot 0 \cdot 0$$
$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$
$$= \max((594026 + 0 + 4 \cdot 0 / 1150) / 1754, 0 / 1754)$$
$$= 338.705 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$
$$= 2 \cdot 48 + 6 \cdot 174 / (0 + 0.5)$$
$$= 2178.000 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$
$$= 1315 \cdot \sin(\pi / 36)$$
$$= 114.610 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{115}{2 \cdot 48 + 174}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	338.705	483.465	
Radial Distance between Hub and Bolts:	62.000	67.500	
Radial Distance between Bolts and Edge:	50.000	57.500	
Circ. Spacing between the Bolts:	109.300	114.610	2178.000

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 483$$

$$= 847909.56 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 594026 - 594026$$

$$= 0.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1315 - 1150) / 2$$

$$= 82.6275 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (67.5 + 38.5 + 82.6) / 2$$

$$= 94.3137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 67.5 + (38.5 / 2)$$

$$= 86.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	546705.	86.7500	1.0000	47427.
Face Pressure, Mt	47321.	94.3137	1.0000	4463.
Gasket Load, Mg	0.	82.6275	1.0000	0.
Gasket Seating, Matm	847910.	82.6275	1.0000	70061.

Total Moment for Operation, Mop 51890. kgf-m.
 Total Moment for Gasket seating, Matm 70061. kgf-m.

Effective Hub Length, $h_o = \sqrt{Bcor \cdot goCor}$ 160.998 mm
 Hub Ratio, $h/h_o = HL / H_o$ 0.323
 Thickness Ratio, $g1/g_o = (g1Cor/goCor)$ 1.638

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.872
 Factor V per 4.16.5 0.338
 Factor f per 4.16.5 1.317

Factors from Table 4-16.4 K = 1.296
 T = 1.799 U = 8.377
 Y = 7.623 Z = 3.938
 e = 0.0054 1/mm d = 0.22067E+07 mm³
 Stress Factors Alpha = 1.940
 Beta = 2.254 Gamma = 1.079



Delta = 2.367 Lambda = 3.445

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.32 \cdot 51890 / 1103) / (3.45 \cdot 38.5^2)$$

$$= 1213.32 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.32 \cdot 70061 / 1103) / (3.45 \cdot 38.5^2)$$

$$= 1638.21 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.25 \cdot 51890 / 1103) / (3.45 \cdot 174^2)$$

$$= 102.23 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.25 \cdot 70061 / 1103) / (3.45 \cdot 174^2)$$

$$= 138.03 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.62 \cdot 51890 / (174^2 \cdot 1103)) - 3.94 \cdot 102$$

$$= 788.80 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (7.62 \cdot 70061 / (174^2 \cdot 1103)) - 3.94 \cdot 138$$

$$= 1065.03 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1213 + \max(102, 789)) / 2$$

$$= 1001.06 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (1638 + \max(138, 1065)) / 2$$

$$= 1351.62 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 594026 / 483$$

$$= 1228.69 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (0 / 483)$$

$$= 0.00 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1213.32	2043.45	1638.21	2523.66
Radial Flange	102.23	1362.30	138.03	1682.44
Tangential Flange	788.80	1362.30	1065.03	1682.44
Maximum Average	1001.06	1362.30	1351.62	1682.44
Bolting	1228.69	1753.82	0.00	1753.82

Minimum Required Flange Thickness 154.229 mm

Estimated M.A.W.P. (Operating) 59.9 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 936.0 kg
 Estimated Unfinished Weight of Forging at given Thk 1153.6 kg

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 70061 / 1 \cdot 100000 \cdot 0.34 / (2.63 \cdot 2067025 \cdot 23.5^2 \cdot 161 \cdot 0.3)$
 $= 0.850$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 51890 / 1 \cdot 100000 \cdot 0.34 / (2.63 \cdot 1909280 \cdot 23.5^2 \cdot 161 \cdot 0.3)$
 $= 0.681$ (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 70061 / 1 \cdot 100000 \cdot 0.34 / (3.45 \cdot 2067025 \cdot 23.5^2 \cdot 161 \cdot 0.3)$
 $= 0.649$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 51890 / 1 \cdot 100000 \cdot 0.34 / (3.45 \cdot 1909280 \cdot 23.5^2 \cdot 161 \cdot 0.3)$
 $= 0.520$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 3 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.12 : 11 °C
 The computed stress ratio is : 0.743

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1183.24	2523.66	1583.31	2523.66
Radial Flange	107.18	1682.44	143.42	1682.44
Tangential Flange	755.01	1682.44	1010.29	1682.44
Maximum Average	969.13	1682.44	1296.80	1682.44
Bolting	1228.69	1753.82	0.00	1753.82

Minimum Required Flange Thickness, no Corrosion: 149.454 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Ch Cover		163.32	303	1.5	1680	1488.7	1
Ch Flg		163.27	328	1.5	1100	1488.7	2
LHS Ch Shell		163.27	66	1.5	1100	1672.3	3
LHS Shell Flg		57.215	183	1.5	1100	1362.3	4
Main Shell		57.216	25	1.5	1100	1435.7	5
Dish End		57.216	32	1.5	1100	1435.7	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Ch Cover		163.158	173	213	295	286.761	1
Ch Flg		163.158	183	213	320	312.598	2
LHS Ch Shell		163.158	185	239	66	58.0607	3
LHS Shell Flg		57.1054	59.8	79.3	175	154.229	4
Main Shell		57.1054	59.8	79.3	25	23.9218	5
Dish End		57.1054	70	92.5	26	21.4551	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

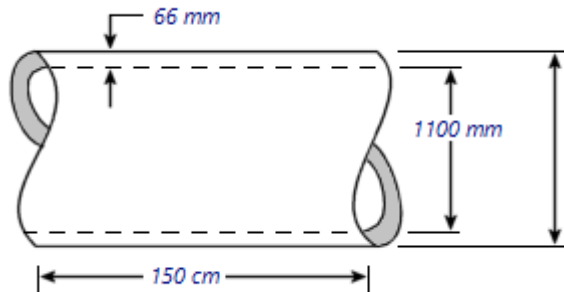
Shell Side MAWP = 58.9 kgf/cm²
 Shell Side MAPnc = 79.3 kgf/cm²
 Channel Side MAWP = 172 kgf/cm²
 Channel Side MAPnc = 175 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2 Class 2, 2023



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	163.269 kgf/cm²	Int. Design Temperature	300 °C	
Material		SA-387 11	Ext. Design Temperature	300 °C	
Length		150.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1672.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	1100.000 mm	Actual Stress	1396.0 kgf/cm²	
Spec. Min. Thk t _e , or sa		66.0000 mm	Spec. Nominal Thk.	66.000 mm	
Surface Area		58056.6 cm²	SG of Contents	1.000	
Element Volume		1425497.6 cm³	Weight of Contents	1424.9 kgf	
Empty Weight		2810.6 kgf	Operating Weight	4235.5 kgf	
			Comp. All. Ope. Stress	1014.9 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1103 \cdot (\exp(163/(1672 \cdot 1)) - 1) + 1.5 + 0$
 $= 56.6 + 1.5 + 0 = 58.1 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1672 \cdot 1 \cdot \ln((2 \cdot 64.5 + 1103) / 1103) - 0.11$
 $= 185 - 0.11 = 185 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2111 \cdot 1 \cdot \ln((2 \cdot 66 + 1100) / 1100)$
 $= 239 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 163 \cdot 1103 / (1(1232 - 1103))$
 $= 1396.0095 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 163 / (1 \cdot \ln((2 \cdot 64.5 + 1103) / 1103))$
 $= 1476.1393 \text{ kgf/cm}^2$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.660 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

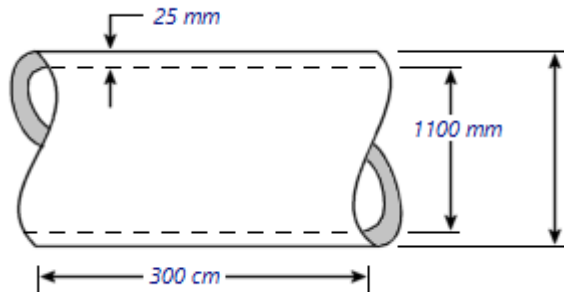
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2 Class 2, 2023



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	57.216 kgf/cm²	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	270 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1435.7 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	1100.000 mm	Actual Stress	1342.7 kgf/cm²	
Spec. Min. Thk t _e , or sa		25.0000 mm	Spec. Nominal Thk.	25.000 mm	
Surface Area		108384.9 cm²	SG of Contents	1.000	
Element Volume		1625345.8 cm³	Weight of Contents	1624.6 kgf	
Empty Weight		2054.4 kgf	Operating Weight	3679.0 kgf	
			Comp. All. Ope. Stress	930.4 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1103 \cdot (\exp(57.2/(1436 \cdot 1)) - 1) + 1.5 + 0$$

$$= 22.4 + 1.5 + 0 = 23.9 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1436 \cdot 1 \cdot \ln((2 \cdot 23.5 + 1103) / 1103) - 0.11$$

$$= 59.9 - 0.11 = 59.8 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784 \cdot 1 \cdot \ln((2 \cdot 25 + 1100) / 1100)$$

$$= 79.3 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 57.2 \cdot 1103 / (1 (1150 - 1103))$$

$$= 1342.7410 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 57.2 / (1 \cdot \ln((2 \cdot 23.5 + 1103) / 1103))$$

$$= 1371.1462 \text{ kgf/cm}^2$$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.222 %

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 5 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.12 : 9 °C
 The computed stress ratio is : 0.788

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (18.5 \cdot 1) / (25 - 1.5 - 0)$$

$$= 0.788$$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	58.912 kgf/cm ²
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	25.000 mm	Basic MDMT	14 °C
Stress/Thk. Ratio	0.788	Temperature Difference	9 °C
		Computed Min. Temp.	5 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 3142 / \sqrt{(575 \cdot 23.5)}$$

$$= 27.034$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, Ch = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 27 - 1.058$$

$$= 0.0342$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.034 \cdot 1909280 (23.5 / 1150)$$

$$= 2136.115 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 2136.1 / 2153.6$$

$$= 0.992$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2154 / 4921$$

$$= 0.438$$

Ratio [AE]:

$$= Fhe / E$$

$$= 2136 / 1909280$$

$$= 0.001119$$



Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.44) + (0.20000\text{E-}04 - 0.002)) / \ln(\ln(1 + 0.20000\text{E-}04) / \ln(1 + 0.002))$$

$$= 0.180$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys}) m1$$

$$= 2154 (1 + 0.0020) / \ln(1 + 0.0020) 0.17993604$$

$$= 6603.125 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1 - 0.44)$$

$$= 0.337$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (4921 \cdot \exp(0.34)) / (0.34 0.34)$$

$$= 9950.704 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.441.5 - 0.5 \cdot 0.442.5 - 0.443.5$$

$$= 0.315$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.296	0.468E-05	0.280E-05	0.958E-05	-0.353E-05	0.711E+05	0.335E-01
2	1230	-4.115	0.199E-06	0.199E-06	0.246E-05	-0.245E-05	0.108E+07	0.192E-04
3	615	-5.524	0.844E-08	0.844E-08	0.630E-06	-0.630E-06	0.185E+07	0.786E-03
4	922	-4.819	0.536E-07	0.536E-07	0.140E-05	-0.140E-05	0.158E+07	0.536E-03
5	1076	-4.467	0.108E-06	0.108E-06	0.189E-05	-0.189E-05	0.135E+07	0.321E-03
6	1153	-4.291	0.148E-06	0.148E-06	0.216E-05	-0.216E-05	0.122E+07	0.170E-03
7	1191	-4.203	0.172E-06	0.172E-06	0.231E-05	-0.231E-05	0.115E+07	0.811E-04
8	1211	-4.159	0.185E-06	0.185E-06	0.238E-05	-0.238E-05	0.111E+07	0.324E-04
9	1220	-4.137	0.192E-06	0.192E-06	0.242E-05	-0.242E-05	0.110E+07	0.698E-05
10	1225	-4.126	0.195E-06	0.195E-06	0.244E-05	-0.243E-05	0.109E+07	0.602E-05
11	1223	-4.131	0.194E-06	0.193E-06	0.243E-05	-0.243E-05	0.109E+07	0.506E-06
12	1224	-4.129	0.194E-06	0.194E-06	0.243E-05	-0.243E-05	0.109E+07	0.275E-05
13	1223	-4.130	0.194E-06	0.194E-06	0.243E-05	-0.243E-05	0.109E+07	0.112E-05
14	1223	-4.131	0.194E-06	0.194E-06	0.243E-05	-0.243E-05	0.109E+07	0.308E-06
15	1223	-4.131	0.194E-06	0.193E-06	0.243E-05	-0.243E-05	0.109E+07	0.991E-07

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 1223/2154$$

$$= 1.986$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1223/1.99$$

$$= 615.942 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

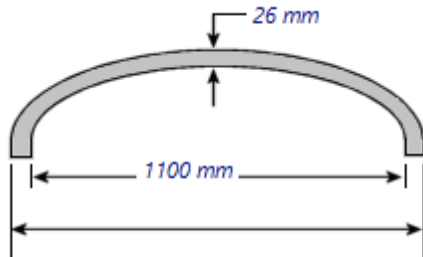
$$= 2 \cdot Fha(t / D_o)$$

$$= 2 \cdot 616(23.5/1150)$$

$$= 25.173 \text{ kgf/cm}^2$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	57.216 kgf/cm²	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	270 °C	
Length		5.080 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1435.7 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	1100.000 mm	Actual Stress	1172.0 kgf/cm²	
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.	32.000 mm	
Surface Area		16544.5 cm²	SG of Contents	1.000	
Element Volume		222504.4 cm³	Weight of Contents	222.4 kgf	
Empty Weight		425.2 kgf	Operating Weight	647.6 kgf	
			Comp. All. Ope. Stress	930.4 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 20 mm

t = tr + ci + co
 = 20 + 1.5 + 0
 = 21.5 mm - see below for the derivation

For Elliptical heads: Compute [h, k, r.] per 4.3.7.1

h = D/(2 • Aspect Ratio)
 = 1100/(2 • 2)
 = 275.000 mm

k = D/(2 • h)
 = 1100/(2 • 275)
 = 2.000

r = D(0.5/k - 0.08) + ca
 = 1100 (0.5/2 - 0.08) + 1.5
 = 188.500 mm



$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 1100(0.44 \cdot 2 + 0.02) + 1.5 \\ &= 991.500 \text{ mm} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 20 \text{ mm}$, $P = 57.2 \text{ kgf/cm}^2$

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 1103 - 189)/(991 - 189)] \\ &= 1.1 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(991 \cdot 20) / 189} \\ &= 0.75 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 1103 - 189) / \cos(1.1 - 0.75) + 189 \\ &= 576 \text{ mm - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:
 $r/D > 0.08$, Thus:
 $C1 = 0.692(r/D) + 0.605$
 $= 0.692(189/1103) + 0.605$
 $= 0.72$ - Equation 4.3.13

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(189/1103) \\ &= 1.02, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:
 $= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$
 $= 0.72 \cdot 1909280 \cdot 20^2 / (1.02 \cdot 576(0.5 \cdot 576 - 189))$
 $= 9465 \text{ kgf/cm}^2$, Equation 4.3.16

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:
 $C3 = S_y = 2154 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 2154 \cdot 20 / (1.02 \cdot 576(0.5 \cdot 576 / 189 - 1)) \\ &= 139 \text{ kgf/cm}^2, \text{ Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:
 $G = P_{eth}/P_y = 9465/139 = 67.9$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 67.9 - 0.20354 \cdot 67.9^2 + 0.019274 \cdot 67.9^3) / \\ &\quad (1 + 0.19014 \cdot 67.9 - 0.089534 \cdot 67.9^2 + 0.0093965 \cdot 67.9^3)) \cdot 139 \\ &= 282 \text{ kgf/cm}^2, \text{ Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:
 $= P_{ck}/1.5 = 282/1.5 = 188 \text{ kgf/cm}^2$ - Equation 4.3.21

Step 9: Compute allowable pressure based on rupture of crown [P_{ac}]:
 $= 2 \cdot S \cdot E / (L/t + 0.5)$
 $= 2 \cdot 1436 \cdot 1 / (991/20 + 0.5)$
 $= 57.2 \text{ kgf/cm}^2$, Equation 4.3.22

Step 10: Compute final pressure [P_a]:
 $= \min(P_{ak}, P_{ac}) = \min(188, 57.2)$
 $= 57.2 \text{ kgf/cm}^2$, Equation 4.3.23: (same as original design pressure)

Allowable pressure in Crown at given thickness [P_{ac}]:



$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1436 \cdot 1 / (991 / 24.5 + 0.5) \\ &= 70.088 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:
= MaxPressure - Pliq = 70.1 - 0.11 = 70 kgf/cm²

Computed Maximum Allowable Pressure New and Cold [MAPnc]:
= 92.5 kgf/cm²

Computed Membrane Stress Intensity at Design Pressure [Sm]:
= 1172.0 kgf/cm²

Required Thickness of Straight Flange = 23.922 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.823 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 27.721 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
= square root($4 \cdot 16545 / 3.14159$)
= 1451.384 mm

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 1 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.12 : 13 °C
The computed stress ratio is : 0.674

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(16.5 \cdot 1) / (26 - 1.5 - 0)$
= 0.674

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 3 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 19 °C
Temperature reduction per Graph 3.12 : 16 °C
The computed stress ratio is : 0.607

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(18.5 \cdot 1) / (32 - 1.5 - 0)$
= 0.607

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	58.912 kgf/cm ²
Governing Thk.	32.000 mm	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.607	Basic MDMT	19 °C
		Temperature Difference	16 °C
		Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:



Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

EXTERNAL PRESSURE RESULTS for: Dish End

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y (t / R_o) \\ &= 0.075 \cdot 1909280 (24.5/993) \\ &= 3532.024 \text{ kgf/cm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 3532.0 / 2153.6 \\ &= 1.640 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_{ys} / \sigma_{uts} \\ &= 2154/4921 \\ &= 0.438 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 3532/1909280 \\ &= 0.001850 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} &= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})) \\ &= (\ln(0.44) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002)) \\ &= 0.180 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} &= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1} \\ &= 2154 (1 + 0.0020) / \ln(1 + 0.0020)^{0.17993604} \\ &= 6603.125 \text{ kgf/cm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} &= m2f(m2f2 - R) \\ &= 0.6 (1 - 0.44) \\ &= 0.337 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} &= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2}) \\ &= (4921 \cdot \exp(0.34)) / (0.34^{0.34}) \\ &= 9950.704 \text{ kgf/cm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} &= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5 \\ &= 1.5 \cdot 0.441.5 - 0.5 \cdot 0.442.5 - 0.443.5 \\ &= 0.315 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.296	0.468E-05	0.280E-05	0.958E-05	-0.353E-05	0.711E+05	0.328E-01
2	1230	-4.115	0.199E-06	0.199E-06	0.246E-05	-0.245E-05	0.108E+07	0.712E-03
3	1845	-2.706	0.126E-05	0.123E-05	0.545E-05	-0.526E-05	0.312E+06	0.407E-02
4	1537	-3.410	0.550E-06	0.547E-06	0.381E-05	-0.378E-05	0.607E+06	0.685E-03
5	1384	-3.762	0.340E-06	0.339E-06	0.310E-05	-0.309E-05	0.824E+06	0.170E-03
6	1461	-3.586	0.435E-06	0.434E-06	0.344E-05	-0.343E-05	0.709E+06	0.210E-03
7	1422	-3.674	0.385E-06	0.384E-06	0.327E-05	-0.325E-05	0.765E+06	0.898E-05
8	1403	-3.718	0.362E-06	0.361E-06	0.318E-05	-0.317E-05	0.794E+06	0.831E-04
9	1413	-3.696	0.374E-06	0.373E-06	0.322E-05	-0.321E-05	0.780E+06	0.377E-04
10	1417	-3.685	0.379E-06	0.378E-06	0.325E-05	-0.323E-05	0.772E+06	0.145E-04
11	1420	-3.680	0.382E-06	0.381E-06	0.326E-05	-0.324E-05	0.769E+06	0.283E-05



12	1421	-3.677	0.384E-06	0.383E-06	0.326E-05	-0.325E-05	0.767E+06	0.306E-05
13	1420	-3.679	0.383E-06	0.382E-06	0.326E-05	-0.325E-05	0.768E+06	0.116E-06
14	1420	-3.679	0.383E-06	0.382E-06	0.326E-05	-0.325E-05	0.768E+06	0.136E-05
15	1420	-3.679	0.383E-06	0.382E-06	0.326E-05	-0.325E-05	0.768E+06	0.621E-06
16	1420	-3.679	0.383E-06	0.382E-06	0.326E-05	-0.325E-05	0.768E+06	0.252E-06
17	1420	-3.679	0.383E-06	0.382E-06	0.326E-05	-0.325E-05	0.768E+06	0.684E-07

Factor of Safety [FS]:

Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$
 $= 2.407 - 0.741 \cdot F_{ic} / F_y$
 $= 2.407 - 0.741 \cdot 1421/2154$
 $= 1.918$

Allowable Circumferential Compressive Stress [Fha]:

$= F_{ic}/FS$
 $= 1421/1.92$
 $= 740.706 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot F_{ha}[t/R_o]$
 $= 2 \cdot 741 [24.5/993]$
 $= 36.540 \text{ kgf/cm}^2$

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Hydrostatic Test Pressure Results:

Horizontal Test performed per:

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	1786.0	3002.9	0.595	197.55
Main Shell	1626.2	2518.6	0.646	67.86
Dish End	1390.1	2518.6	0.552	67.86

Theoretical Maximum Allowable test pressure Shellside is : 104.985 kgf/cm²

Theoretical Maximum Allowable test pressure Tubeside : 332.027 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
N2-Tube Outlet	Nozzle	1407.13	1407.13	1.000
N3-Shell Inlet	Nozzle	1203.20	1203.20	1.000
N3-Shell Inlet	Pad	1407.13	1398.99	1.006
N4-Shell outlet	Nozzle	1203.20	1203.20	1.000
N4-Shell outlet	Pad	1407.13	1398.99	1.006
K1-Vent	Nozzle	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Ch Cover	1876.18	1488.73	1.260
Ch Flg	1876.18	1488.73	1.260
LHS Ch Shell	2110.70	1672.27	1.262
LHS Shell Flg	1682.44	1362.30	1.235
Main Shell	1784.41	1435.72	1.243
Dish End	1784.41	1435.72	1.243

Minimum 1.235

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1111.45	1.083
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	1475.61	1829.27	0.807
N2-Tube Outlet	1475.61	1829.27	0.807
N3-Shell Inlet	568.12	1564.16	0.363
N4-Shell outlet	568.12	1564.16	0.363
K1-Vent	214.19	1564.16	0.137

Elements Suitable for Test Pressure.

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FileName : 401-E-31004-AK

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Pressure Test Results: Shop Test Step: 10 5:03pm May 21,2026



HEXAGON

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2 Class 2, 2023

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	293.5	No Calc	No Calc
20	30	No Calc	...	318.5	No Calc	No Calc
30	40	150	1232	64.5	No Calc	No Calc
40	50	No Calc	...	173.5	No Calc	No Calc
50	60	314.247	1150	23.5	No Calc	No Calc
60	70	No Calc	1152	24.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	295	199.34	...	No Calc
20	30	320	312.598	...	No Calc
30	40	66	No Calc	...	146
40	50	175	154.229	1.05461	No Calc
50	60	25	6.89256	1.05461	25.2
60	70	26	4.20117	1.05461	36.5

Minimum

25.2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	150	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	314.247	2539997	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	5205.63	...	5205.63	...	520.563
20	30	3406.66	408445	3389.44	408964	340.666
30	40	2810.64	1425497	2750.3	1433283	281.064
40	50	976.633	224278	967.179	224548	97.6633
50	60	2054.4	1625345	1933.71	1640918	205.44
60	70	425.175	222504	405.245	224197	42.5175
Total		14879	3906070.00	14651	3931910.50	1487



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	...	14.75	2	...	Liquid: 10
10	Insl	47.4897	14.5	...	...	Hot Insulation
20	Liqd	408.261	21	...	...	Liquid: 20
20	Insl	30.9555	20.5	...	...	Hot Insulation
30	Liqd	1424.86	75	...	...	Liquid: 30
30	Insl	99.3467	75	...	...	Hot Insulation
30	Nozl	636.333	75	80.4	...	N1-Tube Inlet
30	Nozl	705.413	75	-80.4	...	N2-Tube Outlet
40	Liqd	224.177	11.35	...	...	Liquid: 40
40	Insl	21.5762	11.35	...	...	Hot Insulation
50	Sadl	407.405	75	75.45	...	Left Saddle
50	Sadl	407.405	265	75.45	...	New Saddle
50	Liqd	1624.62	150	...	...	Liquid: 50
50	Insl	265.767	150	...	...	Hot Insulation
50	Nozl	63.4645	30	-62.3165	...	N3-Shell Inlet
50	Nozl	63.4645	30	62.3165	...	N4-Shell outlet
50	Nozl	9.48338	250	57.0472	...	K1-Vent
60	Liqd	222.404	14.2467	...	...	Liquid: 60
60	Insl	52.9217	16.29	...	...	Hot Insulation
30	FTsh	2341.99	161.5	...	...	
30	Tube	5208.93	280	...	...	Tubes
40	Bafl	46.6085	52.6	...	...	Baffles
40	Bafl	46.6085	97.6	...	...	Baffles
40	Bafl	46.6085	142.6	...	...	Baffles
40	Bafl	46.6085	187.6	...	...	Baffles
40	Bafl	47.0331	232.6	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	814.8
Liquid	3904.3
Insulation	518.1
Nozzles	1478.2
Exchanger Components	7784.4
Liquid in Tubes	767.3
Sum of the Detail Weights	15267.0 kg

Weight Summation Results: (kg)

Fabricated	Shop Test	Shipping	Erected	Empty	Operating
------------	-----------	----------	---------	-------	-----------



Main Elements	16367.1	16367.1	16367.1	16367.1	16367.1	16367.1
Saddles	814.8	814.8	814.8	814.8	814.8	814.8
Nozzles	1478.2	1478.2	1478.2	1478.2	1478.2	1478.2
Exchanger	7784.4	7784.4	7784.4	7784.4	7784.4	7784.4
Insulation	...	...	518.1	518.1	518.1	518.1
Ope. Liquid	...	...	...	...	...	3904.3
Tube Ope Lqd	...	...	...	...	...	767.3
Test Liquid	...	3904.3	...	...	...	...
Tube Tst Lqd	...	767.3	...	...	...	...
Totals	26444.4	31116.0	26962.5	26962.5	26962.5	31634.0

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	26444.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	31116.0 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	26962.5 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	26962.5 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	26962.5 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	31634.0 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	31383.6 kg

Exchanger Tube Data

Volume of Exchanger tubes :	767597.6 cm ³
Weight of Ope Liq in tubes :	767.3 kg
Weight of Water in tubes :	767.3 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	38159	...
20	30	33845.6	14790.6
30	40	58056.6	51836.3
40	50	16706.2	8121.02
50	60	108385	103673
60	70	16544.5	14871.7

Total	271697.0	193292.2 cm ²
Total	292.5	208.1 ft ²



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3-Shell Inlet	83.80	104.12	270	600	GR 1.1	42.593	83.111	41.210	83.111
N4-Shell outlet	83.80	104.12	270	600	GR 1.1	42.593	83.111	41.210	83.111
K1-Vent	83.80	104.12	270	600	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 83.111 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 104.115 kgf/cm²

Channelside Flange Rating

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N3-Shell Inlet	192.151	0.50	569	510	58.802	83.111	Pass
N4-Shell outlet	192.151	0.50	569	510	58.912	83.111	Pass

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(569) + 4 \cdot \text{abs}(510) 192.151 \leq \pi \cdot 192.151^3 [(83.802 - 58.802) + .500 \cdot 83.802]$$

$$= 0.94932 + 04 \leq 0.14911 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(83.802, 0.500 \cdot 83.802 + 83.802 - 9493.2 / (\pi \cdot 192.151^3))$$

$$= \min(83.802, 83.111)$$

$$= 83.111 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N4-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(569) + 4 \cdot \text{abs}(510) 192.151 \leq \pi \cdot 192.151^3 [(83.802 - 58.912) + .500 \cdot 83.802]$$

$$= 0.94932 + 04 \leq 0.14887 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(83.802, 0.500 \cdot 83.802 + 83.802 - 9493.2 / (\pi \cdot 192.151^3))$$

$$= \min(83.802, 83.111)$$

$$= 83.111 \text{ kgf/cm}^2$$

Flange Pressure Ratings are per ASME B16.5 2020 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 4.450

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.99 • 1 • 0.6 • 57.5²)

= 1968.9 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.67 • 0.78 • 1969

= 1027.7 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	1.30	1.0000	1.0000	1.0000	57.50	0.7817
20	1.30	1.0000	1.0000	1.0000	57.50	0.7817
30	1.30	1.0000	1.0000	1.0000	57.50	0.7817
40	1.30	1.0000	1.0000	1.0000	57.50	0.7817
50	1.30	1.0000	1.0000	1.0000	57.50	0.7817
60	1.30	1.0000	1.0000	1.0000	57.50	0.7817

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	130	226.373	6678	200.853	115.285
20	30	130	156.6	6577.2	200.853	113.545
30	40	130	173.04	25956	200.853	448.09
40	50	130	166.8	3786.36	200.853	65.3657
50	60	130	172.8	51840	200.853	894.938
60	70	130	173.04	5930.19	200.853	102.376

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.





Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1) \cdot 1$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 3821 \cdot 0.094 \cdot 1$
 $= 358.6 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.094 \cdot 30569 \cdot 1$
 $= 1912.7 \text{ kgf}$

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
10	20	84	3821.1	358.634
20	30	55	3821.1	358.634
30	40	55	3821.1	358.634
40	50	55	3821.1	358.634
50	Sad1	55	3821.1	358.634
Sad1	60	55	3821.1	358.634
50	60	55	3821.1	358.634
60	70	55	3821.1	358.634

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	438.788 cm
Center of Gravity of the Liquid	281.446 cm
Center of Gravity of the Insulation	317.981 cm
Center of Gravity of the Nozzles	162.830 cm
Center of Gravity of the Tubesheet(s)	233.952 cm
Center of Gravity of the Tubes	352.452 cm
Center of Gravity of the Baffles	388.851 cm
Center of Gravity of Bare Shell New and Cold	136.343 cm
Center of Gravity of Bare Shell Corroded	133.382 cm
Vessel CG in the Operating Condition	215.287 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	202.840 cm
Vessel CG in the Test Condition	216.177 cm

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	562.50	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.80	cm
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1435.72	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		795.42	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		19697.01	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-1.22	-2518.70	
Long. Stress at Top of Midspan	1.22	1435.72	
Long. Stress at Bottom of Midspan	1.22	1435.72	
Long. Stress at Top of Saddles	148.98	1435.72	
Long. Stress at Bottom of Saddles	-82.57	-2518.70	
Long. Stress at Bottom of Saddles	82.57	1435.72	
Tangential Shear in Shell	79.61	1147.86	
Circ. Stress at Horn of Saddle	163.38	2153.58	
Circ. Compressive Stress in Shell	10.90	1435.72	

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 795 • 1.5 • 1000/1900
 = 628.0 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 19069 + max(628, 0, 0, 0)
 = 19697.0 kgf

Summary of Loads at the Base of this Saddle:



Vertical Load (including saddle weight)	20104.42	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	1193.13	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 1000.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h m^2)}{3L} \right) \right]$$

$$= -19697 \cdot 75 \left[1 - \left(1 - \frac{75}{328} + \frac{(56.2^2 - 28.8^2)}{(2 \cdot 75 \cdot 328)} \right) / \left(1 + \frac{(4 \cdot 28.8)}{3 \cdot 328} \right) \right]$$

$$= -3947.0 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(1 + 2 \frac{(R_m^2 - h m^2)}{(L^2)} \right) / \left(1 + \frac{(4h m^2)}{3L} \right) - 4a/L$$

$$= 19697 \cdot 328 / 4 \left(1 + 2 \frac{(56.2^2 - 28.8^2)}{(328^2)} \right) / \left(1 + \frac{(4 \cdot 28.8)}{3 \cdot 328} \right) - 4 \cdot 75 / 328$$

$$= 303.2 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 0 \cdot 562 / (2 \cdot 25) - 303 / (\pi \cdot 562^2 \cdot 25)$$

$$= -1.22 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 562 / (2 \cdot 25) + 303 / (\pi \cdot 562^2 \cdot 25)$$

$$= 1.22 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 562 / (2 \cdot 25) - 3947 / (0.11 \cdot \pi \cdot 562^2 \cdot 25)$$

$$= 148.98 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 562 / (2 \cdot 25) + 3947 / (0.19 \cdot \pi \cdot 562^2 \cdot 25)$$

$$= -82.57 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 19697 (328 - 2 \cdot 75) / (328 + (4 \cdot 28.8 / 3))$$

$$= 9562.9 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 9563 / (562 \cdot 25)$$

$$= 79.61 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(562 \cdot 25)}$$

$$= 92.497 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(562 \cdot 25)}, 2 \cdot 75)$$



= 334.99 mm

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 = $-K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 = $-0.76 \cdot 19697 \cdot 0.1 / (335(25 + 1 \cdot 16))$
 = -10.90 kgf/cm^2

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:
 = $-Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$
 = $-19697 / (4(25 + 1 \cdot 16)335) -$
 $12 \cdot 0.053 \cdot 19697 \cdot 562 / (328(25 + 1 \cdot 16)^2)$
 = -163.38 kgf/cm^2

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$
 = $((1.56 \cdot \sqrt{(550 \cdot 25) + 380})^3) / 12$
 = 73.298 cm^4

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = $73.3 + 141(158 - 12.5)^2$
 = 29938.061 cm^4

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	562.9	25.0	12.5	140.7	175914.4	0.299E+05
Wearplate	380.0	16.0	33.0	60.8	200640.0	0.954E+04
Web	30.0	369.0	225.5	110.7	2496284.5	0.176E+05
BasePlate	190.0	40.0	430.0	76.0	3267999.5	0.563E+05
Totals	...	...	...	388.2	6140838.5	0.113E+06

Distance to Centroid [C1]:
 = AY / A
 = $6140838 / 388$
 = 158.175 mm

Angle [β]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 120 / 2$
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = $K1 \cdot Q$
 = $0.2 \cdot 19697$
 = 4008.7690 kgf



$$\begin{aligned} \text{Tension Stress, } S_t &= (F_h / A_s) = 16.1971 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } S_a &= 0.6 * \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Dimension [d]:} \\ &= B - R * \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000 - 550 * \sin(120/2) / 1.05 \\ &= 545.154 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = F_h * d = 2185.3975 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } S_b &= (M * C_1 / I) = 30.5061 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } S_a &= 2/3 * \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))} \\ &= \sqrt{(3(19697 + 407)190 / (4 * 1000 * 1436))} \\ &= 14.126 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned} \text{Web Length Dimension [Web Length]:} \\ &= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 * \cos(90 - 120/2) (550 + 25 + 16) \\ &= 1023.642 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Distance between Ribs [e]:} \\ &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 1024 / (5 - 1) \\ &= 255.910 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Pressure Area [Ap]:} \\ &= e * \text{Bpwid} / 2 \\ &= 256 * 190 / 2 \\ &= 243.115 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Bearing Pressure [Bp]:} \\ &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth}) \\ &= 20104 / (1000 * 190) \\ &= 10.581 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Axial Load [P]:} \\ &= A_p * B_p \\ &= 243 * 10.6 \\ &= 2572.466 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Area of the Rib and Web [Ar]:} \\ &= \text{Rib Area} + \text{Web Area} \\ &= 42 + 38.4 \\ &= 80.387 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Compressive Stress [Sc]:} \\ &= P / A_r \\ &= 2572 / 80.4 \\ &= 32.001 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

$$\begin{aligned} \text{Rib dimension [D]:} \\ &= \text{Saddle Width} - \text{Web Thickness} \\ &= 150 - 30 \\ &= 120.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Distance to Centroid from Datum [ytot]:} \\ &= AY / A \end{aligned}$$



$$= 0/80.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(984/80.4)}$$

$$= 34.994 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2)(550 + 25 + 16) - 40$$

$$= 664.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2)(550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1000 - 50.8$$

$$= 949.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1024 - 949) / 2$$

$$= 37.221 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(664^2 + 37.2^2)}$$

$$= 665.542 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038899 / 2154)}$$

$$= 136.704$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 666 / 35$$

$$= 19.019$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1193 / (2 \cdot 1000) \cdot 256 \cdot 666 / 2$$

$$= 50.803 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 32 / 1241 + (50.8 \cdot 75 / 9843750) / 1436$$

$$= 0.053$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.



Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(591/2)(40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034/119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1193/(2 \cdot 1000)$$

$$= 0.597 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.6 \cdot 256 \cdot 369$$

$$= 56.334 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 56.3/137877$$

$$= 40.858 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5)^3 / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 41.5/1261 + (56.3 \cdot 75/1034) / 1436$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	20104.4	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1193.1	kgf
Maximum Transverse Force	Ft	0.0	kgf
Maximum Longitudinal Force	Fl	1193.1	kgf
Saddle Bolted to Steel Foundation	No		



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1193 / (4.23 \cdot 4)$
= 70.4 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $0 / (4.23 \cdot 4)$
= 0.0 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 ($QO > QL \rightarrow$ No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $1193 / (1100 \cdot 4)$
= 0.2713 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1000 \cdot 0 + 0$
= 0.00 kgf-m .

Eccentricity (e):
= $Rmom / QO$
= $0 / 20104$
= $0.00 \text{ mm} < Bplen/6 \rightarrow$ No Uplift in Transverse direction

Bolt Area due to Transverse Load [$Bltareart$]:
= 0 (No Uplift)

Required Area of a Single Bolt [$Bltarear$]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.27, 0]$
= 0.2713 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.88	cm
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1435.72	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		795.42	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		19571.81	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-1.27	-2518.70	
Long. Stress at Top of Midspan	1.27	1435.72	
Long. Stress at Bottom of Midspan	1.27	1435.72	
Long. Stress at Top of Saddles	157.14	1435.72	
Long. Stress at Bottom of Saddles	-87.09	-2518.70	
Long. Stress at Bottom of Saddles	87.09	1435.72	
Tangential Shear in Shell	84.02	1147.86	
Circ. Stress at Horn of Saddle	174.31	2153.58	
Circ. Compressive Stress in Shell	11.43	1435.72	

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 795 • 1.5 • 1000/1900
 = 628.0 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 18944 + max(628, 0, 0, 0)
 = 19571.8 kgf

Summary of Loads at the Base of this Saddle:



Vertical Load (including saddle weight)	19979.22	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	1193.13	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 1000.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h m^2)}{3L} \right) \right]$$

$$= -19572 \cdot 75 \left[1 - \left(1 - \frac{75}{328} + \frac{(56.3^2 - 28.9^2)}{(2 \cdot 75 \cdot 328)} \right) / \left(1 + \frac{(4 \cdot 28.9)}{3 \cdot 328} \right) \right]$$

$$= -3923.7 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(1 + 2 \frac{(R_m^2 - h m^2)}{(L^2)} \right) / \left(1 + \frac{(4h m^2)}{3L} \right) - 4a/L$$

$$= 19572 \cdot 328 / 4 \left(1 + 2 \frac{(56.3^2 - 28.9^2)}{(328^2)} \right) / \left(1 + \frac{(4 \cdot 28.9)}{3 \cdot 328} \right) - 4 \cdot 75 / 328$$

$$= 298.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 0 \cdot 563 / (2 \cdot 23.5) - 298 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= -1.27 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 563 / (2 \cdot 23.5) + 298 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= 1.27 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 563 / (2 \cdot 23.5) - 3924 / (0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 157.14 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 563 / (2 \cdot 23.5) + 3924 / (0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= -87.09 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 19572 (328 - 2 \cdot 75) / (328 + (4 \cdot 28.9 / 3))$$

$$= 9499.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 9499 / (563 \cdot 23.5)$$

$$= 84.02 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$



= 329.48 mm

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 = $-K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 = $-0.76 \cdot 19572 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$
 = -11.43 kgf/cm^2

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:
 = $-Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$
 = $-19572 / (4(23.5 + 1 \cdot 16)329) -$
 $12 \cdot 0.053 \cdot 19572 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$
 = -174.31 kgf/cm^2

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 = $((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)^3) / 12$
 = 60.303 cm^4

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = $60.3 + 131(161 - 11.8)^2$
 = 29118.506 cm^4

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:
 = AY / A
 = $6081764 / 379$
 = 160.666 mm

Angle [β]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 120 / 2$
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = $K1 \cdot Q$
 = $0.2 \cdot 19572$
 = 3983.2876 kgf



$$\begin{aligned} \text{Tension Stress, } S_t &= (F_h / A_s) = 16.0941 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } S_a &= 0.6 * \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Dimension [d]:} \\ &= B - R * \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000 - 552 * \sin(120/2) / 1.05 \\ &= 543.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = F_h * d = 2166.5652 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } S_b &= (M * C_1 / I) = 31.3890 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } S_a &= 2/3 * \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))} \\ &= \sqrt{(3(19572 + 407)190 / (4 * 1000 * 1436))} \\ &= 14.082 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned} \text{Web Length Dimension [Web Length]:} \\ &= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 * \cos(90 - 120/2) (550 + 25 + 16) \\ &= 1023.642 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Distance between Ribs [e]:} \\ &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1024 / (5 - 1) \\ &= 255.910 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Pressure Area [Ap]:} \\ &= e * \text{Bpwid} / 2 \\ &= 256 * 190 / 2 \\ &= 243.115 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Bearing Pressure [Bp]:} \\ &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth}) \\ &= 19979 / (1000 * 190) \\ &= 10.515 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Axial Load [P]:} \\ &= A_p * B_p \\ &= 243 * 10.5 \\ &= 2556.445 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Area of the Rib and Web [Ar]:} \\ &= \text{Rib Area} + \text{Web Area} \\ &= 42 + 38.4 \\ &= 80.387 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Compressive Stress [Sc]:} \\ &= P / A_r \\ &= 2556 / 80.4 \\ &= 31.802 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

$$\begin{aligned} \text{Rib dimension [D]:} \\ &= \text{Saddle Width} - \text{Web Thickness} \\ &= 150 - 30 \\ &= 120.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Distance to Centroid from Datum [ytot]:} \\ &= AY / A \end{aligned}$$



$$= 0/80.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(984/80.4)}$$

$$= 34.994 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2)(550 + 25 + 16) - 40$$

$$= 664.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2)(550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1000 - 50.8$$

$$= 949.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1024 - 949) / 2$$

$$= 37.221 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(664^2 + 37.2^2)}$$

$$= 665.542 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038899 / 2154)}$$

$$= 136.704$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 666 / 35$$

$$= 19.019$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1193 / (2 \cdot 1000) \cdot 256 \cdot 666 / 2$$

$$= 50.803 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/r < C_c$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot C1 / I) / \text{Sba}$$

$$= 31.8 / 1241 + (50.8 \cdot 75 / 9843750) / 1436$$

$$= 0.053$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.



Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(591/2)(40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034/119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1193/(2 \cdot 1000)$$

$$= 0.597 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.6 \cdot 256 \cdot 369$$

$$= 56.334 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 56.3/137877$$

$$= 40.858 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 /$$

$$(5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5)^3 / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 41.3/1261 + (56.3 \cdot 75/1034) / 1436$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	19979.2	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1193.1	kgf
Maximum Transverse Force	Ft	0.0	kgf
Maximum Longitudinal Force	Fl	1193.1	kgf
Saddle Bolted to Steel Foundation	No		



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1193 / (4.23 \cdot 4)$
= 70.4 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $0 / (4.23 \cdot 4)$
= 0.0 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 ($QO > QL \rightarrow$ No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot Nbolts)$
= $1193 / (1100 \cdot 4)$
= 0.2713 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1000 \cdot 0 + 0$
= 0.00 kgf-m .

Eccentricity (e):
= $Rmom / QO$
= $0 / 19979$
= $0.00 \text{ mm} < Bplen/6 \rightarrow$ No Uplift in Transverse direction

Bolt Area due to Transverse Load [$Bltareart$]:
= 0 (No Uplift)

Required Area of a Single Bolt [$Bltarear$]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.27, 0]$
= 0.2713 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1362.30	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		57461.47	kgf
Pressure used in Analysis	P	57.160	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	656.10	1435.72
Long. Stress at Bottom of Midspan	713.92	1435.72
Long. Stress at Top of Saddles	968.56	1435.72
Long. Stress at Bottom of Saddles	527.86	1435.72
Tangential Shear in Shell	275.65	1147.86
Circ. Stress at Horn of Saddle	511.75	1794.65
Circ. Compressive Stress in Shell	33.57	1435.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1740/2 + 0) \cdot 1000/1000$
 $= 2609.4 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(462, 0) \cdot 1000/1900$
 $= 243.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(2869, 0) \cdot 1000/1900$
 $= 1510.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2869/2 + 0) \cdot 1000/1000$
 $= 4303.6 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$



$$= 53158 + \max(243, 2609, 1510, 4304)$$

$$= 57461.5 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1970 (0 + \pi/4(1.36 \cdot 1.2)^2)$$

$$= 4532.336 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	57868.88	kgf
Transverse Shear Load	1434.53	kgf
Longitudinal Shear Load	2869.07	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1000.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -57461 \cdot 75 [1 - (1 - 75/328 + (56.3^2 - 0^2)/(2 \cdot 75 \cdot 328))/(1 + (4 \cdot 0)/(3 \cdot 328))]$$

$$= -7080.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 57461 \cdot 328/4(1 + 2(56.3^2 - 0^2)/(328^2))/(1 + (4 \cdot 0)/(3 \cdot 328)) - 4 \cdot 75/328$$

$$= 6771.5 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 57.2 \cdot 563/(2 \cdot 23.5) - 6771/(\pi \cdot 563^2 \cdot 23.5)$$

$$= 656.10 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 57.2 \cdot 563/(2 \cdot 23.5) + 6771/(\pi \cdot 563^2 \cdot 23.5)$$

$$= 713.92 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 57.2 \cdot 563/(2 \cdot 23.5) - 7080/(0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 968.56 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 57.2 \cdot 563/(2 \cdot 23.5) + 7080/(0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 527.86 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 57461 (328 - 2 \cdot 75)/(328 + (4 \cdot 0/3))$$

$$= 31165.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 31166/(563 \cdot 23.5)$$

$$= 275.65 \text{ kgf/cm}^2$$



Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1436/1436, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57461 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$$

$$= -33.57 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -57461 / (4(23.5 + 1 \cdot 16)329) - 12 \cdot 0.053 \cdot 57461 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$$

$$= -511.75 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 1900.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13139\text{E-}04 \cdot 1900(270 - 21.1)$$

$$= 6.213 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)23.5^3)/12$$

$$= 60.303 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 60.3 + 131(161 - 11.8)^2$$

$$= 29118.506 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:

$$= AY / A$$



$$= 6081764/379$$

$$= 160.666 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 57461$$

$$= 11694.6562 \text{ kgf}$$

Tension Stress, $St = (Fh/As) = 47.2512 \text{ kgf/cm}^2$

Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 552 \cdot \sin(120/2) / 1.05$$

$$= 543.913 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 6360.8853 \text{ kgf-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 92.1560 \text{ kgf/cm}^2$

Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(57461 + 407)190) / (4 \cdot 1000 \cdot 1436)}$$

$$= 23.966 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1024 / (5 - 1)$$

$$= 255.910 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 256 \cdot 190/2$$

$$= 243.115 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 57869 / (1000 \cdot 190)$$

$$= 30.457 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 243 \cdot 30.5$$

$$= 7404.626 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 42 + 38.4$$

$$= 80.387 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$



$$= 7405/80.4$$

$$= 92.113 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150 - 30$$

$$= 120.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/80.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(984/80.4)}$$

$$= 34.994 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (550 + 25 + 16) - 40$$

$$= 664.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1000 - 50.8$$

$$= 949.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1024 - 949) / 2$$

$$= 37.221 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{664^2 + 37.2^2}$$

$$= 665.542 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2154)}$$

$$= 136.704$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 666/35$$

$$= 19.019$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 2869 / (2 \cdot 1000) \cdot 256 \cdot 666/2$$

$$= 122.164 \text{ kgf-m.}$$



Compressive Allowable, $KL/r < C_c$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 92.1 / 1241 + (122 \cdot 75 / 9843750) / 1436$$

$$= 0.139$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150 / 2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(591/2) (40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034 / 119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 369 / 29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 2869 / (2 \cdot 1000)$$

$$= 1.435 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.43 \cdot 256 \cdot 369$$

$$= 135.465 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 135 / 137877$$

$$= 98.250 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5^3) / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 120 / 1261 + (135 \cdot 75 / 1034) / 1436$$



= 0.163

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load QO (Weight)	QO	53565.3	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1510.0	kgf
Maximum Transverse Force	Ft	1434.5	kgf
Maximum Longitudinal Force	Fl	2869.1	kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2869 / (4.23 \cdot 4)$$

$$= 169.4 \text{ kgf/cm}^2. \text{ Must be less than } 1099.6 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1435 / (4.23 \cdot 4)$$

$$= 84.7 \text{ kgf/cm}^2. \text{ Must be less than } 1099.6 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2869 / (1100 \cdot 4)$$

$$= 0.6523 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 1435 + 0$$

$$= 1434.54 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1435 / 53565$$

$$= 26.78 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.65, 0]$$

$$= 0.6523 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	6 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	6 °C
Temperature reduction per Graph 3.12	:	0 °C
The computed stress ratio is	:	1.000



The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (18.5 \cdot 1) / (16 - 1.5 - 0)$$

$$= 1.000$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.88	cm
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1435.72	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		26077.84	kgf
Pressure used in Analysis	P	57.160	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	683.32	1435.72
Long. Stress at Bottom of Midspan	686.71	1435.72
Long. Stress at Top of Saddles	894.38	1435.72
Long. Stress at Bottom of Saddles	568.97	1435.72
Tangential Shear in Shell	111.95	1147.86
Circ. Stress at Horn of Saddle	232.25	1794.65
Circ. Compressive Stress in Shell	15.23	1435.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (1740/2 + 0) \cdot 1000/1000$$

$$= 2609.4 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(462, 0) \cdot 1000/1900$$

$$= 243.2 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(2869, 0) \cdot 1000/1900$$

$$= 1510.0 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (2869/2 + 0) \cdot 1000/1000$$



= 4303.6 kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 21774 + max(243, 2609, 1510, 4304)
 = 26077.8 kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1970 (0 + $\pi/4$ (1.36 • 1.2)²)
 = 4532.336 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26485.25	kgf
Transverse Shear Load	1434.53	kgf
Longitudinal Shear Load	2869.07	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 1000.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$
 = $-26078 \cdot 75 [1 - (1 - 75/328 + (56.3^2 - 28.9^2)/(2 \cdot 75 \cdot 328))/(1 + (4 \cdot 28.9)/(3 \cdot 328))]$
 = -5228.1 kgf-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 (1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L$
 = $26078 \cdot 328/4 (1 + 2(56.3^2 - 28.9^2)/(328^2))/(1 + (4 \cdot 28.9)/(3 \cdot 328)) - 4 \cdot 75/328$
 = 397.5 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 = $57.2 \cdot 563/(2 \cdot 23.5) - 397/(\pi \cdot 563^2 \cdot 23.5)$
 = 683.32 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $57.2 \cdot 563/(2 \cdot 23.5) + 397/(\pi \cdot 563^2 \cdot 23.5)$
 = 686.71 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$
 = $57.2 \cdot 563/(2 \cdot 23.5) - 5228/(0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$
 = 894.38 kgf/cm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

= $P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$
 = $57.2 \cdot 563/(2 \cdot 23.5) + 5228/(0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$
 = 568.97 kgf/cm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$
 = $26078 (328 - 2 \cdot 75)/(328 + (4 \cdot 28.9/3))$
 = 12657.3 kgf



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 12657 / (563 \cdot 23.5)$$

$$= 111.95 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1436/1436, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 26078 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$$

$$= -15.23 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -26078 / (4(23.5 + 1 \cdot 16)329) -$$

$$12 \cdot 0.053 \cdot 26078 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$$

$$= -232.25 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)^3) / 12$$

$$= 60.303 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 60.3 + 131(161 - 11.8)^2$$

$$= 29118.506 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6081764/379$$

$$= 160.666 \text{ mm}$$



Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 26078 \\ &= 5307.4062 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 21.4441 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000 - 552 \cdot \sin(120/2) / 1.05 \\ &= 543.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2886.7712 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 41.8233 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(26078 + 407)190 / (4 \cdot 1000 \cdot 1436)} \\ &= 16.213 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (550 + 25 + 16) \\ &= 1023.642 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1024 / (5 - 1) \\ &= 255.910 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 256 \cdot 190 / 2 \\ &= 243.115 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 26485 / (1000 \cdot 190) \\ &= 13.940 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 243 \cdot 13.9 \\ &= 3388.926 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 42 + 38.4 \\ &= 80.387 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 3389 / 80.4 \\ &= 42.158 \text{ kgf/cm}^2 \end{aligned}$$



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 150 - 30 \\
 &= 120.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/80.4 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 150/2 \\
 &= 75.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(984/80.4)} \\
 &= 34.994 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1000 - \cos(120/2) (550 + 25 + 16) - 40 \\
 &= 664.500 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2) (550 + 25 + 16) \\
 &= 1023.642 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1000 - 50.8 \\
 &= 949.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1024 - 949) / 2 \\
 &= 37.221 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{(Lv^2 + Dd^2)} \\
 &= \sqrt{(664^2 + 37.2^2)} \\
 &= 665.542 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038899/2154)} \\
 &= 136.704
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 666/35 \\
 &= 19.019
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 2869 / (2 \cdot 1000) \cdot 256 \cdot 666/2 \\
 &= 122.164 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$



$$= (1 - (19)^2 / (2 \cdot 137^2)) \cdot 2154 / (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 42.2/1241 + (122 \cdot 75/9843750) / 1436$$

$$= 0.099$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000 - \cos(591/2) (40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034/119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2869 / (2 \cdot 1000)$$

$$= 1.435 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.43 \cdot 256 \cdot 369$$

$$= 135.465 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 135/137877$$

$$= 98.250 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) \cdot 2154 / (5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5^3) / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 54.7/1261 + (135 \cdot 75/1034) / 1436$$

$$= 0.112$$

Input Data for Base Plate Bolting Calculations:



Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load QO (Weight)	QO	22181.6	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1510.0	kgf
Maximum Transverse Force	Ft	1434.5	kgf
Maximum Longitudinal Force	Fl	2869.1	kgf
Saddle Bolted to Steel Foundation		No	

Friction force due to thermal growth, [Ff]:
 = 16069.6 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 16070 / (4.23 • 4)
 = 948.6 kgf/cm². Must be less than 1099.6 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 1435 / (4.23 • 4)
 = 84.7 kgf/cm². Must be less than 1099.6 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 2869 / (1100 • 4)
 = 0.6523 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1000 • 1435 + 0
 = 1434.54 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 1435 / 22182
 = 64.67 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.65, 0]
 = 0.6523 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 6 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 6 °C
 Temperature reduction per Graph 3.12 : 0 °C
 The computed stress ratio is : 1.000



The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (18.5 \cdot 1) / (16 - 1.5 - 0)$$

$$= 1.000$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Shell Allowable Stress used in Calculation		2518.70	kgf/cm ²
Head Allowable Stress used in Calculation		2518.70	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	21703.46	kgf
Pressure used in Analysis		67.805	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	801.66	2518.70
Long. Stress at Bottom of Midspan	823.50	2518.70
Long. Stress at Top of Saddles	919.67	2518.70
Long. Stress at Bottom of Saddles	753.22	2518.70
Tangential Shear in Shell	104.11	1413.13
Circ. Stress at Horn of Saddle	193.29	3778.06
Circ. Compressive Stress in Shell	12.68	2518.70

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (574/2 + 0) \cdot 1000/1000$
 $= 861.1 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(153, 0) \cdot 1000/1900$
 $= 80.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 20842 + \max(80.3, 861, 0, 0)$
 $= 21703.5 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1970 (0 + \pi/4 (1.36 \cdot 1.2)^2)$
 $= 4532.336 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22110.87	kgf
---	----------	-----



Transverse Shear Load 287.03 kgf
 Longitudinal Shear Load 152.52 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 1000.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm^2)}{(3L)} \right) \right]$$

$$= -21703 \cdot 75 \left[1 - \left(1 - \frac{75}{328} + \frac{(56.3^2 - 0^2)}{(2 \cdot 75 \cdot 328)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 328)} \right) \right]$$

$$= -2674.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(1 + 2 \frac{(Rm^2 - hm^2)}{(L^2)} \right) / \left(1 + \frac{(4hm^2)}{(3L)} \right) - 4a/L$$

$$= 21703 \cdot 328 / 4 \left(1 + 2 \frac{(56.3^2 - 0^2)}{(328^2)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 328)} \right) - 4 \cdot 75 / 328$$

$$= 2557.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) - 2558 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= 801.66 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) + 2558 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= 823.50 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) - 2674 / (0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 919.67 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) + 2674 / (0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 753.22 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 21703 (328 - 2 \cdot 75) / (328 + (4 \cdot 0 / 3))$$

$$= 11771.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 11771 / (563 \cdot 23.5)$$

$$= 104.11 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$



Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_{6,r}]:
 = $-K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 = $-0.76 \cdot 21703 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$
 = -12.68 kgf/cm²

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_{7,r*}]:
 = $-Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$
 = $-21703 / (4(23.5 + 1 \cdot 16)329) - 12 \cdot 0.053 \cdot 21703 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$
 = -193.29 kgf/cm²

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$
 = $((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)23.5^3) / 12$
 = 60.303 cm⁴

Resolved Inertia for the Shell [I_{l,shell}]:
 = ShellInertia + AShell(C₁ - Y)²
 = 60.3 + 131 (161 - 11.8)²
 = 29118.506 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C₁]:
 = AY / A
 = 6081764/379
 = 160.666 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K₁]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 = $(1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 = 0.2035

Saddle Splitting Force [F_h]:
 = K₁ • Q
 = 0.2 • 21703
 = 4417.1250 kgf

Tension Stress, St = (F_h/A_s) = 17.8470 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1292.1501 kgf/cm²



Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 552 \cdot \sin(120/2) / 1.05$$

$$= 543.913 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 2402.5352 \text{ kgf-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 34.8077 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(21703 + 407) \cdot 190 / (4 \cdot 1000 \cdot 1436)}$$

$$= 14.814 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1024 / (5 - 1)$$

$$= 255.910 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 256 \cdot 190 / 2$$

$$= 243.115 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22111 / (1000 \cdot 190)$$

$$= 11.637 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 243 \cdot 11.6$$

$$= 2829.201 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 42 + 38.4$$

$$= 80.387 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2829 / 80.4$$

$$= 35.195 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150 - 30$$

$$= 120.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 80.4$$

$$= 0.000 \text{ mm}$$



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(984/80.4)}$$

$$= 34.994 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2)(550 + 25 + 16) - 40$$

$$= 664.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2)(550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1000 - 50.8$$

$$= 949.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd})/2$$

$$= (1024 - 949)/2$$

$$= 37.221 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(664^2 + 37.2^2)}$$

$$= 665.542 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2154)}$$

$$= 136.704$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 666/35$$

$$= 19.019$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 153 / (2 \cdot 1000) \cdot 256 \cdot 666/2$$

$$= 6.494 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I})/\text{Sba}$$

$$= 35.2/1241 + (6.49 \cdot 75/9843750)/1436$$

$$= 0.032$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(591/2)(40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034/119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 153/(2 \cdot 1000)$$

$$= 0.076 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.076 \cdot 256 \cdot 369$$

$$= 7.201 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 7.2/137877$$

$$= 5.223 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 /$$

$$(5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5)^3 / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 45.7/1261 + (7.2 \cdot 75/1034) / 1436$$

$$= 0.040$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	21249.8	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	80.3	kgf
Maximum Transverse Force	Ft	287.0	kgf
Maximum Longitudinal Force	F1	152.5	kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $153 / (4.23 \cdot 4)$
= 9.0 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $287 / (4.23 \cdot 4)$
= 16.9 kgf/cm^2 . Must be less than 1099.6 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $153 / (1100 \cdot 4)$
= 0.0347 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1000 \cdot 287 + 0$
= 287.03 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $287 / 21250$
= $13.51 \text{ mm} < B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.035, 0]$
= 0.0347 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.88	cm
Shell Allowable Stress used in Calculation		2518.70	kgf/cm ²
Head Allowable Stress used in Calculation		2518.70	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 21700.24 kgf



Pressure used in Analysis P 67.805 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	811.17	2518.70
Long. Stress at Bottom of Midspan	813.99	2518.70
Long. Stress at Top of Saddles	986.80	2518.70
Long. Stress at Bottom of Saddles	716.01	2518.70
Tangential Shear in Shell	93.16	1413.13
Circ. Stress at Horn of Saddle	193.26	3778.06
Circ. Compressive Stress in Shell	12.68	2518.70

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (574/2 + 0) \cdot 1000/1000$
 $= 861.1\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(153, 0) \cdot 1000/1900$
 $= 80.3\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 20839 + \max(80.3, 861, 0, 0)$
 $= 21700.2\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1970 (0 + \pi/4(1.36 \cdot 1.2)^2)$
 $= 4532.336\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22107.65	kgf
Transverse Shear Load	287.03	kgf
Longitudinal Shear Load	152.52	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 1000.000\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)]$
 $= -21700 \cdot 75 [1 - (1 - 75/328 + (56.3^2 - 28.9^2) / (2 \cdot 75 \cdot 328)) / (1 + (4 \cdot 28.9) / (3 \cdot 328))]$
 $= -4350.4\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L$
 $= 21700 \cdot 328 / 4 (1 + 2(56.3^2 - 28.9^2)/(328^2)) / (1 + (4 \cdot 28.9) / (3 \cdot 328)) - 4 \cdot 75/328$
 $= 330.8\ kgf \cdot m.$



Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) - 331 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= 811.17 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) + 331 / (\pi \cdot 563^2 \cdot 23.5)$$

$$= 813.99 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) - 4350 / (0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 986.80 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 67.8 \cdot 563 / (2 \cdot 23.5) + 4350 / (0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$$

$$= 716.01 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 21700 (328 - 2 \cdot 75) / (328 + (4 \cdot 28.9 / 3))$$

$$= 10532.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 10533 / (563 \cdot 23.5)$$

$$= 93.16 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 21700 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$$

$$= -12.68 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -21700 / (4(23.5 + 1 \cdot 16)329) -$$

$$12 \cdot 0.053 \cdot 21700 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$$

$$= -193.26 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:



$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)^3) / 12$$

$$= 60.303 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 60.3 + 131 (161 - 11.8)^2$$

$$= 29118.506 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6081764 / 379$$

$$= 160.666 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 21700$$

$$= 4416.4702 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 17.8444 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000 - 552 \cdot \sin(120 / 2) / 1.05$$

$$= 543.913 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2402.1790 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 34.8026 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(21700 + 407) \cdot 190) / (4 \cdot 1000 \cdot 1436)}$$

$$= 14.813 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1024 / (5 - 1)$$

$$= 255.910 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



$$= 256 \cdot 190/2$$

$$= 243.115 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22108 / (1000 \cdot 190)$$

$$= 11.636 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 243 \cdot 11.6$$

$$= 2828.790 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 42 + 38.4$$

$$= 80.387 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2829 / 80.4$$

$$= 35.190 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150 - 30$$

$$= 120.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 80.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150 / 2$$

$$= 75.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(984 / 80.4)}$$

$$= 34.994 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (550 + 25 + 16) - 40$$

$$= 664.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (550 + 25 + 16)$$

$$= 1023.642 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1000 - 50.8$$

$$= 949.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (1024 - 949) / 2$$

$$= 37.221 \text{ mm}$$



Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(664^2 + 37.2^2)}$$

$$= 665.542 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038899 / 2154)}$$

$$= 136.704$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 666 / 35$$

$$= 19.019$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 153 / (2 \cdot 1000) \cdot 256 \cdot 666 / 2$$

$$= 6.494 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 137^2)) 2154 /$$

$$(5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3))$$

$$= 1241 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 35.2 / 1241 + (6.49 \cdot 75 / 9843750) / 1436$$

$$= 0.032$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150 / 2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(591/2) (40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034 / 119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369 / 29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 153 / (2 \cdot 1000)$$

$$= 0.076 \text{ kgf/mm}$$



Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.076 \cdot 256 \cdot 369$$

$$= 7.201 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 7.2/137877$$

$$= 5.223 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5)^3 / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 45.7/1261 + (7.2 \cdot 75/1034) / 1436$$

$$= 0.040$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1398.98	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load QO (Weight)	QO	21246.5	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	80.3	kgf
Maximum Transverse Force	Ft	287.0	kgf
Maximum Longitudinal Force	Fl	152.5	kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 153 / (4.23 \cdot 4)$$

$$= 9.0 \text{ kgf/cm}^2. \text{ Must be less than } 1099.6 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 287 / (4.23 \cdot 4)$$

$$= 16.9 \text{ kgf/cm}^2. \text{ Must be less than } 1099.6 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 153 / (1100 \cdot 4)$$

$$= 0.0347 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000 \cdot 287 + 0$$

$$= 287.03 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 287/21247$$

$$= 13.51 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$



Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.035, 0]
= 0.0347 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1362.30	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		57461.47	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-41.55	-1938.23
Long. Stress at Top of Midspan		41.55	1435.72
Long. Stress at Bottom of Midspan		16.27	1435.72
Long. Stress at Top of Saddles		270.90	1435.72
Long. Stress at Bottom of Saddles		-169.80	-1938.23
Long. Stress at Bottom of Saddles		169.80	1435.72
Tangential Shear in Shell		275.65	1147.86
Circ. Stress at Horn of Saddle		511.75	1794.65
Circ. Compressive Stress in Shell		33.57	1435.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1740/2 + 0) \cdot 1000/1000$
 $= 2609.4 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(462, 0) \cdot 1000/1900$
 $= 243.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2869, 0) \cdot 1000/1900$
 $= 1510.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2869/2 + 0) \cdot 1000/1000$
 $= 4303.6 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned} &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\ &= 53158 + \max(243, 2609, 1510, 4304) \\ &= 57461.5 \text{ kgf} \end{aligned}$$

Longitudinal Wind Force [FI]:

$$\begin{aligned} &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.1 \cdot 1970 (0 + \pi/4 (1.36 \cdot 1.2)^2) \\ &= 4532.336 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	57868.88	kgf
Transverse Shear Load	1434.53	kgf
Longitudinal Shear Load	2869.07	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned} &= \text{Baseplate Length} \\ &= 1000.000 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)] \\ &= -57461 \cdot 75 [1 - (1 - 75/328 + (56.3^2 - 0^2) / (2 \cdot 75 \cdot 328)) / (1 + (4 \cdot 0)/(3 \cdot 328))] \\ &= -7080.1 \text{ kgf-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/3L) - 4a/L \\ &= 57461 \cdot 328/4 (1 + 2(56.3^2 - 0^2)/(328^2)) / (1 + (4 \cdot 0)/(3 \cdot 328)) - 4 \cdot 75/328 \\ &= 6771.5 \text{ kgf-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\ &= -1.05 \cdot 563/(2 \cdot 23.5) - 6771/(\pi \cdot 563^2 \cdot 23.5) \\ &= -41.55 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\ &= -1.05 \cdot 563/(2 \cdot 23.5) + 6771/(\pi \cdot 563^2 \cdot 23.5) \\ &= 16.27 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\ &= -1.05 \cdot 563/(2 \cdot 23.5) - 7080/(0.11 \cdot \pi \cdot 563^2 \cdot 23.5) \\ &= 270.90 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t) \\ &= -1.05 \cdot 563/(2 \cdot 23.5) + 7080/(0.19 \cdot \pi \cdot 563^2 \cdot 23.5) \\ &= -169.80 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\ &= 57461 (328 - 2 \cdot 75)/(328 + (4 \cdot 0/3)) \\ &= 31165.7 \text{ kgf} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



$$= 1.17 \cdot 31166 / (563 \cdot 23.5)$$

$$= 275.65 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1436/1436, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57461 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$$

$$= -33.57 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -57461 / (4(23.5 + 1 \cdot 16)329) - 12 \cdot 0.053 \cdot 57461 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$$

$$= -511.75 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)^3) / 12$$

$$= 60.303 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 60.3 + 131(161 - 11.8)^2$$

$$= 29118.506 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6081764/379$$

$$= 160.666 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$



$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 57461 \\ &= 11694.6562 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 47.2512 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000 - 552 \cdot \sin(120/2) / 1.05 \\ &= 543.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6360.8853 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 92.1560 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt[3]{(3(57461 + 407) \cdot 190) / (4 \cdot 1000 \cdot 1436)} \\ &= 23.966 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (550 + 25 + 16) \\ &= 1023.642 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 1024 / (5 - 1) \\ &= 255.910 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 256 \cdot 190 / 2 \\ &= 243.115 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 57869 / (1000 \cdot 190) \\ &= 30.457 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 243 \cdot 30.5 \\ &= 7404.626 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 42 + 38.4 \\ &= 80.387 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 7405 / 80.4 \\ &= 92.113 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



	B	D	Y	A	AY	Io
Rib+Web	35.0	150.0	...	52.5	...	984.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 150 - 30 \\
 &= 120.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/80.4 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 150/2 \\
 &= 75.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(984/80.4)} \\
 &= 34.994 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1000 - \cos(120/2) (550 + 25 + 16) - 40 \\
 &= 664.500 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2) (550 + 25 + 16) \\
 &= 1023.642 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1000 - 50.8 \\
 &= 949.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1024 - 949) / 2 \\
 &= 37.221 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{664^2 + 37.2^2} \\
 &= 665.542 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2154)} \\
 &= 136.704
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 666/35 \\
 &= 19.019
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 2869 / (2 \cdot 1000) \cdot 256 \cdot 666/2 \\
 &= 122.164 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($19 < 137$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / \\
 &\quad (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3)) \\
 &= 1241 \text{ kgf/cm}^2
 \end{aligned}$$



AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 92.1/1241 + (122 \cdot 75/9843750) / 1436$$

$$= 0.139$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(591/2) (40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1034/119) }$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2869 / (2 \cdot 1000)$$

$$= 1.435 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.43 \cdot 256 \cdot 369$$

$$= 135.465 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 135/137877$$

$$= 98.250 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 /$$

$$(5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5^3) / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 120/1261 + (135 \cdot 75/1034) / 1436$$

$$= 0.163$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



Vessel Mean Radius	Rm	563.25	mm
Shell Thickness used in this Case	t	25.000	mm
Stiffened Vessel Length per 4.15.6	L	327.78	cm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1435.72	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.88	cm
Shell Allowable Stress used in Calculation		1435.72	kgf/cm ²
Head Allowable Stress used in Calculation		1435.72	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		26077.84	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-14.34	-1938.23
Long. Stress at Top of Midspan	14.34	1435.72
Long. Stress at Bottom of Midspan	-10.94	-1938.23
Long. Stress at Bottom of Midspan	10.94	1435.72
Long. Stress at Top of Saddles	196.73	1435.72
Long. Stress at Bottom of Saddles	-128.69	-1938.23
Long. Stress at Bottom of Saddles	128.69	1435.72
Tangential Shear in Shell	111.95	1147.86
Circ. Stress at Horn of Saddle	232.25	1794.65
Circ. Compressive Stress in Shell	15.23	1435.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1740/2 + 0) \cdot 1000/1000$
 $= 2609.4\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= max(462, 0) \cdot 1000/1900$
 $= 243.2\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= max(2869, 0) \cdot 1000/1900$
 $= 1510.0\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (2869/2 + 0) \cdot 1000/1000$
 $= 4303.6\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + max(Fwl, Fwt, Fsl, Fst)$
 $= 21774 + max(243, 2609, 1510, 4304)$
 $= 26077.8\ kgf$



Longitudinal Wind Force [F_L]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1970 (0 + \pi/4 (1.36 \cdot 1.2)^2)$
 $= 4532.336 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26485.25	kgf
Transverse Shear Load	1434.53	kgf
Longitudinal Shear Load	2869.07	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 1000.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)]$
 $= -26078 \cdot 75 [1 - (1 - 75/328 + (56.3^2 - 28.9^2) / (2 \cdot 75 \cdot 328)) / (1 + (4 \cdot 28.9) / (3 \cdot 328))]$
 $= -5228.1 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/3L) - 4a/L$
 $= 26078 \cdot 328/4 (1 + 2(56.3^2 - 28.9^2)/(328^2)) / (1 + (4 \cdot 28.9) / (3 \cdot 328)) - 4 \cdot 75/328$
 $= 397.5 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 563/(2 \cdot 23.5) - 397/(\pi \cdot 563^2 \cdot 23.5)$
 $= -14.34 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 563/(2 \cdot 23.5) + 397/(\pi \cdot 563^2 \cdot 23.5)$
 $= -10.94 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 563/(2 \cdot 23.5) - 5228/(0.11 \cdot \pi \cdot 563^2 \cdot 23.5)$
 $= 196.73 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= -1.05 \cdot 563/(2 \cdot 23.5) + 5228/(0.19 \cdot \pi \cdot 563^2 \cdot 23.5)$
 $= -128.69 \text{ kgf/cm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot h_m/3))$
 $= 26078 (328 - 2 \cdot 75) / (328 + (4 \cdot 28.9/3))$
 $= 12657.3 \text{ kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (R_m \cdot t)$
 $= 1.17 \cdot 12657 / (563 \cdot 23.5)$
 $= 111.95 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x_1, x_2]:



$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(563 \cdot 23.5)}$$

$$= 89.739 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150 + 1.56 \cdot \sqrt{(563 \cdot 23.5)}, 2 \cdot 75)$$

$$= 329.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1436/1436, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 26078 \cdot 0.1 / (329(23.5 + 1 \cdot 16))$$

$$= -15.23 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -26078 / (4(23.5 + 1 \cdot 16)329) -$$

$$12 \cdot 0.053 \cdot 26078 \cdot 563 / (328(23.5 + 1 \cdot 16)^2)$$

$$= -232.25 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	190.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	30.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2153.6	kgf
Height of Web at Center	Hw,c	260.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(552 \cdot 23.5)} + 380)^3) / 12$$

$$= 60.303 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 60.3 + 131(161 - 11.8)^2$$

$$= 29118.506 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	557.6	23.5	11.8	131.0	153966.0	0.291E+05
Wearplate	380.0	16.0	31.5	60.8	191520.0	0.102E+05
Web	30.0	369.0	224.0	110.7	2479679.5	0.170E+05
BasePlate	190.0	40.0	428.5	76.0	3256599.2	0.546E+05
Totals	...	...	...	378.5	6081764.5	0.111E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6081764/379$$

$$= 160.666 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:



$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 26078 \\ &= 5307.4062 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 21.4441 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1292.1501 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000 - 552 \cdot \sin(120/2) / 1.05 \\ &= 543.913 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2886.7712 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 41.8233 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1435.7224 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(26078 + 407) \cdot 190) / (4 \cdot 1000 \cdot 1436)} \\ &= 16.213 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (550 + 25 + 16) \\ &= 1023.642 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 1024 / (5 - 1) \\ &= 255.910 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 256 \cdot 190 / 2 \\ &= 243.115 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 26485 / (1000 \cdot 190) \\ &= 13.940 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 243 \cdot 13.9 \\ &= 3388.926 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 42 + 38.4 \\ &= 80.387 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 3389 / 80.4 \\ &= 42.158 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B

D

Y

A

AY

Io



Rib+Web	35.0	150.0	...	52.5	...	984.
---------	------	-------	-----	------	-----	------

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 150 - 30 \\
 &= 120.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/80.4 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 150/2 \\
 &= 75.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(984/80.4)} \\
 &= 34.994 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1000 - \cos(120/2) (550 + 25 + 16) - 40 \\
 &= 664.500 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2) (550 + 25 + 16) \\
 &= 1023.642 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1000 - 50.8 \\
 &= 949.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1024 - 949) / 2 \\
 &= 37.221 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{664^2 + 37.2^2} \\
 &= 665.542 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038899 / 2154)} \\
 &= 136.704
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 666/35 \\
 &= 19.019
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 2869 / (2 \cdot 1000) \cdot 256 \cdot 666/2 \\
 &= 122.164 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($19 < 137$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (19)^2 / (2 \cdot 137^2)) 2154 / \\
 &\quad (5/3 + 3 \cdot (19) / (8 \cdot 137) - (19^3) / (8 \cdot 137^3)) \\
 &= 1241 \text{ kgf/cm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$



$$= 42.2/1241 + (122 \cdot 75/9843750)/1436$$

$$= 0.099$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	120.0	0.0	42.0	0.0	976.
Web	255.9	30.0	0.0	76.8	0.0	57.6
Totals	...	...	...	118.8	...	0.103E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 150/2$$

$$= 75.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(591/2)(40 + 0 + 0) - 0$$

$$= 369.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1034/119)}$$

$$= 29.507 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 369/29.5$$

$$= 12.506$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2869/(2 \cdot 1000)$$

$$= 1.435 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.43 \cdot 256 \cdot 369$$

$$= 135.465 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 135/137877$$

$$= 98.250 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 137^2)) 2154 / (5/3 + 3 \cdot (12.5) / (8 \cdot 137) - (12.5)^3 / (8 \cdot 137^3))$$

$$= 1261 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 54.7/1261 + (135 \cdot 75/1034)/1436$$

$$= 0.112$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1672.27	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	66.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		147.45	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

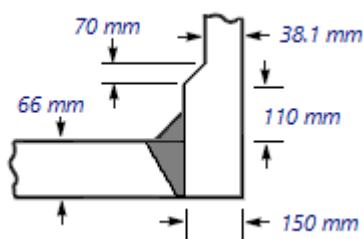
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		508.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	38.1000	mm
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	70.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	508.000	mm.
Actual Thickness Used in Calculation	38.100	mm



Note:



Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
= $0.5 \cdot 511 \cdot (\exp(163/(1407 \cdot 1)) - 1) + 1.5 + 0$
= $31.4 + 1.5 + 0 = 32.9$ mm Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
= $S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
= $1407 \cdot 1 \cdot \ln((2 \cdot 36.6 + 511) / 511) - 0$
= $188 - 0 = 188$ kgf/cm²

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 58.068 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 38.100 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	607.2,	Allowable	:	1407.1 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2910.6 kgf/cm ²	Passed
Occasional	:	531.4,	Allowable	:	1871.5 kgf/cm ²	Passed
Shear	:	54.7,	Allowable	:	985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
= $\min[S_n/S, 1]$
= $\min[1407/1672, 1]$
= 0.841

Thickness of Nozzle at Shell [tn]:
= hub thickness - corrosion allowance
= $150 - 1.5$
= 148.500 mm

Thickness of Nozzle at Top [tn2]:
= thickness - corrosion allowance
= $38.1 - 1.5$
= 36.600 mm

Shell Diameter to Thickness ratio [D/t]:
= D_i/t
= $1103/64.5$
= 17.101 must be less than 400.

Effective Pressure Radius [Reff]:
= $D_i/2 + \text{corrosion allowance}$
= $1100/2 + 1.5$
= 551.500 mm

Effective Length of Vessel Wall [LR]:
= $\min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
= $\min(\sqrt{552 \cdot 64.5}, 2 \cdot 256)$



= 188.605 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 64.5, 0] + \sqrt{(256 \cdot 148)} \\ &= 194.786 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(64.5 + 0) \\ &= 516.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[195, 300, 516] + 64.5 \\ &= 259.286 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 64.5 + ((0 \cdot 1) / 189) \\ &= 64.500 \text{ mm} \end{aligned}$$

Determine Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 256 + 148\right) / \sqrt{\left(\left(1103 + 64.5\right) \cdot 64.5\right)}\right) \\ &= 2.403 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left((\lambda/5)^{0.85}, 1\right) \\ &= 64.5 \cdot 189 \cdot \max\left((2.4/5)^{0.85}, 1\right) \\ &= 121.650 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2C \\ &= 259 + 27.6 \\ &= 286.739 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 122 + 0.84(287 + 0) + 0.5 + 0 + 0 \\ &= 363.427 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 148 / \ln[1 + 148/256] \\ &= 324.100 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 64.5 / \ln[1 + 64.5/552] \\ &= 583.156 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1672 \cdot 1 \\ &= 2508.4 \text{ kgf/cm}^2 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 163 \cdot 324 \cdot 259 \\ &= 137109.4 \text{ kgf} \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 163 \cdot 583 (189 + 148) \\ &= 320743.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163 \cdot 583 \cdot 256 \\ &= 243099.7 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (320744 + 243100 + 137109) / 163 \\ &= 4296.2 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 2508 / (2 \cdot 4296 / 363 - 583 / 64.5) \\ &= 171.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 1672 [64.5 / 583] \\ &= 185.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(172, 185) \\ &= 171.793 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (137109 + 320744 + 243100) / 363 \\ &= 1928.734 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163 \cdot 583 / 64.5 \\ &= 1475.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1929 - 1475, 1475) \\ &= 2382.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2508.41	kgf/cm ²
Local Primary Membrane Stress	PL	2382.32	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	171.79	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (256 + 148) / 256 \\ &= 1.581 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (256 + 148) \\ &= 634.602 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 148.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(243100 \cdot 1.58, 1.5 \cdot 1407(287+0), \pi/4 \cdot 163 \cdot 256^2 \cdot 1.58^2)$$

$$= 209151.766 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 209152 / (635(0.49 \cdot 7.07 + 0.6 \cdot 64.5 + 0.49 \cdot 0))$$

$$= 781.648 < \text{or} = \text{to } 1672.270 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	14 °C
Temperature reduction per Graph 3.13	:	18 °C
The computed stress ratio is	:	0.687

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (44.3 \cdot 1) / (66 - 1.5 - 0)$$

$$= 0.687$$

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 176.7923 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 542.7923 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	66.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1672.27	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-1713.21	kgf
Circumferential Shear	Vc	2549.41	kgf
Longitudinal Shear	Vl	2549.41	kgf
Circumferential Moment	Mc	3822.58	kgf-m.
Longitudinal Moment	Ml	5096.77	kgf-m.
Torsional Moment	Mt	6370.96	kgf-m.



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (18.667) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/D_v (.656) is > 0.5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (5.387) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 808.0 + 2 \cdot 1.65 \cdot \sqrt{(583.75(66.0 - 1.5))}$
 $= 1448.335 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1167.500 mm
 Nozzle Outside Diameter (d) = 808.000 mm
 Vessel Thickness used (T) = 64.500 mm
 Nozzle Thickness used (t) = 148.500 mm
 $T / t = 0.434$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.944$$

Nr/P = 0.060
 Mr/P = 0.106
 M0/P = 0.034
 N0/P = 0.081
 MrD/Mc = 0.263
 NrDL/Mc = 0.098
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.093
 NrDL/Ml = 0.051
 M0D/Ml = 0.025
 N0DL/Ml = 0.090

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	1317	1481	1317	1481
Outplane Stress Summary	1293	1485	1366	1466

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	658	658	658	658



Inplane Stress Summary 595 712 779 558

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		3	3	-3	-3
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		9	9	9	9

Shear Stress Summary 12 12 6 6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		1293	1485	1366	1466

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		3	3	3	3
Outplane Bending (P)		8	-8	8	-8
Outplane Membrane (Mc)		-9	-9	9	9
Outplane Bending (Mc)		-54	54	54	-54
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		658	658	658	658

Outplane Stress Summary 606 698 734 609

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		2	2	2	2
Inplane Bending (P)		26	-26	26	-26
Inplane Membrane (Mc)		-11	-11	11	11
Inplane Bending (Mc)		-179	179	179	-179
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		1317	1481	1317	1481

Inplane Stress Summary 1155 1625 1537 1289

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-3	-3	3	3
Torsional Shear (Mt)		9	9	9	9

Shear Stress Summary 6 6 12 12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	D1
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		1155	1625	1537	1289

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	B1
--------------------	--	----	----	----	----



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	260	456	260	456
Hoop Stress Summary	250	446	277	473

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-11	11	11	-11
Axial Pressure Stress	130	130	130	130
Axial Stress Summary	111	127	157	127

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	2	2

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	250	446	277	473

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	260	456	260	456
Hoop Stress Summary	254	450	273	469

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-27	27	27	-27
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress 130 130 130 130

Axial Stress Summary 98 146 170 108

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	2	2	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	254	450	274	469

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1317	1481	1317	1481	1317	1481	1317	1481
Circ. Pl (SUS)	-10	-10	16	16	-9	-9	13	13
Circ. Q (SUS)	-14	14	30	-30	-153	153	205	-205
Long. Pm (SUS)	658	658	658	658	658	658	658	658
Long. Pl (SUS)	-5	-5	9	9	-6	-6	12	12
Long. Q (SUS)	-58	58	110	-110	-46	46	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	1317	1481	1317	1481	1317	1481	1317	1481
Pm+Pl (SUS)	1307	1471	1333	1497	1308	1472	1330	1494
Pm+Pl+Q (Total)	1293	1485	1363	1467	1155	1625	1535	1289

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1481.00	1672.27	Passed
Pm+Pl (SUS)	1497.01	2508.41	Passed
Pm+Pl+Q (TOTAL)	1625.04	5016.81	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	260	456	260	456	260	456	260	456
Circ. Pl (SUS)	-10	-10	16	16	-6	-6	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	130	130	130	130	130	130	130	130
Long. Pl (SUS)	-11	-11	11	11	-8	-8	8	8
Long. Q (SUS)	-8	8	14	-14	-24	24	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	260.0	456.0	260.0	456.0	260.0	456.0	260.0	456.0
Pm+Pl (SUS)	250.0	446.0	276.0	472.0	254.0	450.0	272.0	468.0
Pm+Pl+Q (Total)	250.2	446.1	276.1	472.0	254.1	450.0	272.2	468.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	456.00	1407.13	Passed
Pm+Pl (SUS)	472.00	2110.70	Passed
Pm+Pl+Q (TOTAL)	472.03	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N2-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	163.268	kgf/cm ²
Temperature for Internal Pressure	Temp	300	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1672.27	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	66.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		147.45	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

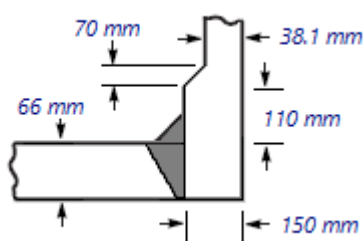
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		508.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	38.1000	mm
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	70.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	508.000	mm.
Actual Thickness Used in Calculation	38.100	mm



Note:



Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
= $0.5 \cdot 511 \cdot (\exp(163/(1407 \cdot 1)) - 1) + 1.5 + 0$
= $31.4 + 1.5 + 0 = 32.9$ mm Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
= $S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
= $1407 \cdot 1 \cdot \ln((2 \cdot 36.6 + 511) / 511) - 0$
= $188 - 0 = 188$ kgf/cm²

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 58.108 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 38.100 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	607.6,	Allowable	:	1407.1 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2910.3 kgf/cm ²	Passed
Occasional	:	531.8,	Allowable	:	1871.5 kgf/cm ²	Passed
Shear	:	54.7,	Allowable	:	985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
= $\min[S_n/S, 1]$
= $\min[1407/1672, 1]$
= 0.841

Thickness of Nozzle at Shell [tn]:
= hub thickness - corrosion allowance
= $150 - 1.5$
= 148.500 mm

Thickness of Nozzle at Top [tn2]:
= thickness - corrosion allowance
= $38.1 - 1.5$
= 36.600 mm

Shell Diameter to Thickness ratio [D/t]:
= D_i/t
= $1103/64.5$
= 17.101 must be less than 400.

Effective Pressure Radius [Reff]:
= $D_i/2 + \text{corrosion allowance}$
= $1100/2 + 1.5$
= 551.500 mm

Effective Length of Vessel Wall [LR]:
= $\min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
= $\min(\sqrt{552 \cdot 64.5}, 2 \cdot 256)$



= 188.605 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 64.5, 0] + \sqrt{(256 \cdot 148)} \\ &= 194.786 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(64.5 + 0) \\ &= 516.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[195, 300, 516] + 64.5 \\ &= 259.286 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 64.5 + ((0 \cdot 1) / 189) \\ &= 64.500 \text{ mm} \end{aligned}$$

Determine Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right) \\ &= \min\left(12, \left(2 \cdot 256 + 148\right) / \left(\sqrt{\left(1103 + 64.5\right) \cdot 64.5}\right)\right) \\ &= 2.403 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 64.5 \cdot 189 \cdot \max((2.4/5)^{0.85}, 1) \\ &= 121.650 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2C \\ &= 259 + 27.6 \\ &= 286.739 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 122 + 0.84(287 + 0) + 0.5 + 0 + 0 \\ &= 363.427 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 148 / \ln[1 + 148/256] \\ &= 324.100 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 64.5 / \ln[1 + 64.5/552] \\ &= 583.156 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1672 \cdot 1 \\ &= 2508.4 \text{ kgf/cm}^2 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 163 \cdot 324 \cdot 259 \\ &= 137201.8 \text{ kgf} \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot RxS(LR + tn) \\ &= 163 \cdot 583 (189 + 148) \\ &= 320959.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot RxS \cdot Rnc \\ &= 163 \cdot 583 \cdot 256 \\ &= 243263.5 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (320960 + 243263 + 137202)/163 \\ &= 4296.2 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - RxS/teff) \\ &= 2508 / (2 \cdot 4296/363 - 583/64.5) \\ &= 171.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/RxS] \\ &= 1672 [64.5/583] \\ &= 185.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(172, 185) \\ &= 171.793 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (137202 + 320960 + 243263)/363 \\ &= 1930.033 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot RxS / teff \\ &= 163 \cdot 583/64.5 \\ &= 1476.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1930 - 1476, 1476) \\ &= 2383.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2508.41	kgf/cm ²
Local Primary Membrane Stress	PL	2383.93	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	171.79	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (256 + 148)/256 \\ &= 1.581 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (256 + 148) \\ &= 634.602 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 148.500, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(243263 \cdot 1.58, 1.5 \cdot 1407(287+0), \pi/4 \cdot 163 \cdot 256^2 \cdot 1.58^2)$
 $= 209292.672 \text{ kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 209293 / (635(0.49 \cdot 7.07 + 0.6 \cdot 64.5 + 0.49 \cdot 0))$
 $= 782.174 < \text{or} = \text{to } 1672.270 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.13 : 17 °C
The computed stress ratio is : 0.688

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (44.4 \cdot 1) / (66 - 1.5 - 0)$
 $= 0.688$

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 176.7923 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 542.7923 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	66.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1672.27	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	163.27	kgf/cm ²
Radial Load	P	-1713.21	kgf
Circumferential Shear	Vc	2549.41	kgf
Longitudinal Shear	Vl	2549.41	kgf
Circumferential Moment	Mc	3822.58	kgf-m.
Longitudinal Moment	Ml	5096.77	kgf-m.
Torsional Moment	Mt	6370.96	kgf-m.



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (18.667) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/D_v (.656) is > 0.5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (5.387) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 808.0 + 2 \cdot 1.65 \cdot \sqrt{(583.75(66.0 - 1.5))}$
 $= 1448.335 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1167.500 mm
 Nozzle Outside Diameter (d) = 808.000 mm
 Vessel Thickness used (T) = 64.500 mm
 Nozzle Thickness used (t) = 148.500 mm
 $T / t = 0.434$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.944$$

Nr/P = 0.060
 Mr/P = 0.106
 M0/P = 0.034
 N0/P = 0.081
 MrD/Mc = 0.263
 NrDL/Mc = 0.098
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.093
 NrDL/Ml = 0.051
 M0D/Ml = 0.025
 N0DL/Ml = 0.090

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	1318	1482	1318	1482
Outplane Stress Summary	1294	1486	1366	1467

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	659	659	659	659



Inplane Stress Summary 595 712 780 559

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		3	3	-3	-3
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		9	9	9	9

Shear Stress Summary 12 12 6 6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		1294	1486	1366	1467

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		3	3	3	3
Outplane Bending (P)		8	-8	8	-8
Outplane Membrane (Mc)		-9	-9	9	9
Outplane Bending (Mc)		-54	54	54	-54
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		659	659	659	659

Outplane Stress Summary 607 698 734 609

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		2	2	2	2
Inplane Bending (P)		26	-26	26	-26
Inplane Membrane (Mc)		-11	-11	11	11
Inplane Bending (Mc)		-179	179	179	-179
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		1318	1482	1318	1482

Inplane Stress Summary 1156 1626 1538 1290

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-3	-3	3	3
Torsional Shear (Mt)		9	9	9	9

Shear Stress Summary 6 6 12 12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 1156 1626 1538 1290

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
--------------------	--	----	----	----	----



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	261	457	261	457
Hoop Stress Summary	250	446	278	474

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-11	11	11	-11
Axial Pressure Stress	130	130	130	130
Axial Stress Summary	111	127	157	127

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	2	2

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	250	446	278	474

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	261	457	261	457
Hoop Stress Summary	254	450	274	470

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-27	27	27	-27
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress 130 130 130 130

Axial Stress Summary 98 146 170 108

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	2	2	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	254	450	274	470

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1318	1482	1318	1482	1318	1482	1318	1482
Circ. Pl (SUS)	-10	-10	16	16	-9	-9	13	13
Circ. Q (SUS)	-14	14	30	-30	-153	153	205	-205
Long. Pm (SUS)	659	659	659	659	659	659	659	659
Long. Pl (SUS)	-5	-5	9	9	-6	-6	12	12
Long. Q (SUS)	-58	58	110	-110	-46	46	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	1318	1482	1318	1482	1318	1482	1318	1482
Pm+Pl (SUS)	1308	1472	1334	1498	1309	1473	1331	1495
Pm+Pl+Q (Total)	1294	1486	1364	1468	1156	1626	1536	1290

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1482.00	1672.27	Passed
Pm+Pl (SUS)	1498.01	2508.41	Passed
Pm+Pl+Q (TOTAL)	1626.04	5016.81	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	261	457	261	457	261	457	261	457
Circ. Pl (SUS)	-10	-10	16	16	-6	-6	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	130	130	130	130	130	130	130	130
Long. Pl (SUS)	-11	-11	11	11	-8	-8	8	8
Long. Q (SUS)	-8	8	14	-14	-24	24	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	261.0	457.0	261.0	457.0	261.0	457.0	261.0	457.0
Pm+Pl (SUS)	251.0	447.0	277.0	473.0	255.0	451.0	273.0	469.0
Pm+Pl+Q (Total)	251.2	447.1	277.1	473.0	255.1	451.0	273.2	469.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	457.00	1407.13	Passed
Pm+Pl (SUS)	473.00	2110.70	Passed
Pm+Pl+Q (TOTAL)	473.03	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N3-Shell Inlet

From: 50

Pressure for Reinforcement Calculations	P	57.215	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1435.72	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	3142.4668	mm
Shell Finished (Minimum) Thickness	t	25.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		298.79	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1398.99	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

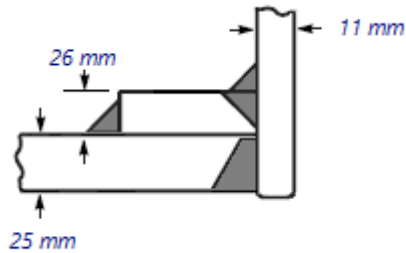
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		146.329	mm.
Actual Thickness Used in Calculation		10.973	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 149 \cdot (\exp(57.2/(1203 \cdot 1)) - 1) + 1.5 + 0$$

$$= 3.64 + 1.5 + 0 = 5.14 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1203 \cdot 1 \cdot \ln((2 \cdot 9.47 + 149) / 149) - 0$$

$$= 144 - 0 = 144 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.720 mm
Min. required shell thickness, internal p + c :	23.925 mm
Min. required shell thickness, external p + c :	8.393 mm

Nozzle neck minimum required thickness : 7.720 mm

Minimum nozzle neck thickness 9.601 mm , is greater than required thickness 7.720 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	615.6,	Allowable :	1203.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	2392.4 kgf/cm ²	Passed
Occasional :	250.1,	Allowable :	1600.3 kgf/cm ²	Passed
Shear :	258.1,	Allowable :	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1203/1436, 1]$$

$$= 0.838$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1399/1436, 1]$$

$$= 0.974$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 11 - 1.5$$

$$= 9.473 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1103/23.5$$

$$= 46.936 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1100/2 + 1.5 \\ &= 551.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned} \text{Limit [LR1]:} \\ &= \sqrt{(R_{eff} \cdot t) + W} \\ &= \sqrt{(552 \cdot 23.5) + 50} \\ &= 163.843 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR2]:} \\ &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((552 + 23.5) \cdot (23.5 + 26))} \\ &= 168.708 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR3]:} \\ &= 2 \cdot R_n \\ &= 2 \cdot 74.7 \\ &= 149.329 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Final value of [LR]:} \\ &= \min(LR1, LR2, LR3) \\ &= \min(164, 169, 149) \\ &= 149.329 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH1]:} \\ &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 23.5, 26] + \sqrt{(74.7 \cdot 9.47)} \\ &= 52.595 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH2]:} \\ &= L_{pr1} \\ &= 300 \\ &= 300.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH3]:} \\ &= 8(t + t_e) \\ &= 8(23.5 + 26) \\ &= 396.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[52.6, 300, 396] + 23.5 \\ &= 76.095 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Vessel Thickness [teff]:} \\ &= t + ((A5 \cdot frp) / LR) \\ &= 23.5 + ((13 \cdot 0.97) / 149) \\ &= 31.983 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Determine Parameter } [\lambda]: \\ &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 74.7 + 9.47) / (\sqrt{(1103 + 32) \cdot 32})) \\ &= 0.833 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned} \text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 23.5 \cdot 149 \cdot \max((0.83/5)^{0.85}, 1) \\ &= 35.092 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot LH \\ &= 9.47 \cdot 76.1 \\ &= 7.208 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 20^2 \\ &= 2.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(50 \cdot 26, 149 \cdot 26) \\ &= 13.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 35.1 + 0.84(7.21 + 0) + 0.5 + 2 + 0 + 0.97(13) \\ &= 56.301 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 9.47 / \ln[1 + 9.47/74.7] \\ &= 79.307 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 32 / \ln[1 + 32/552] \\ &= 567.341 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1436 \cdot 1 \\ &= 2153.6 \text{ kgf/cm}^2 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 57.2 \cdot 79.3 \cdot 76.1 \\ &= 3452.9 \text{ kgf} \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 57.2 \cdot 567 (149 + 9.47) \\ &= 51548.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 57.2 \cdot 567 \cdot 74.7 \\ &= 24236.6 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (51548 + 24237 + 3453) / 57.2 \\ &= 1384.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2154 / (2 \cdot 1385 / 56.3 - 567 / 32) \\ &= 68.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S [t / R_{xs}] \\ &= 1436 [23.5 / 567] \\ &= 59.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(68.5, 59.5) \\ &= 59.469 \text{ kgf/cm}^2 \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (3453 + 51548 + 24237) / 56.3 \\ &= 1407.403 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 57.2 \cdot 567 / 32 \\ &= 1014.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1407 - 1015, 1015) \\ &= 1799.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	1799.87	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	59.47	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (74.7 + 9.47) / 74.7 \\ &= 1.127 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (74.7 + 9.47) \\ &= 132.163 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (74.7 + 9.47 + 50) \\ &= 210.703 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 14.142, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 9.473, 6.000) \text{ Passed} \\ &tf1, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 26.000, 0.6 \cdot 23.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(24237 \cdot 1.13, 1.5 \cdot 1203(7.21+0), \pi/4 \cdot 57.2 \cdot 74.7^2 \cdot 1.13^2) \\ &= 3181.130 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 3181 \cdot 23.5 \cdot 1436 / (23.5 \cdot 1436 + 26 \cdot 1399) \\ &= 1530.807 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 3181 \cdot 26 \cdot 1399 / (23.5 \cdot 1436 + 26 \cdot 1399) \\ &= 1650.323 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 1531 / (132 \cdot (0.6 \cdot 23.5 + 0.49 \cdot 0)) \\ &= 82.147 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 1650 / (132 \cdot (0.6 \cdot 26 + 0.49 \cdot 7.07)) \\ &= 65.498 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned} &= fwp / (L \cdot P \cdot (0.49 \cdot L42T)) \\ &= 1650 / (211 \cdot (0.49 \cdot 14.1)) \\ &= 113.030 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(82.1, 65.5, 113) \\ &= 113.0 \text{ must be less than or equal to } 1435.7 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 9.47 / \ln[1 + 9.47/74.7] \\ &= 79.307 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 32 / \ln[1 + 32/552] \\ &= 567.341 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1436 \\ &= 2153.6 \text{ kgf/cm}^2 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 1.05 \cdot 79.3 \cdot 76.1 \\ &= 63.6 \text{ kgf} \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 1.05 \cdot 567 (149 + 9.47) \\ &= 950.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.05 \cdot 567 \cdot 74.7 \\ &= 446.7 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (950 + 447 + 63.6) / 1.05 \\ &= 1384.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2154 / (2 \cdot 1385 / 56.3 - 567 / 32) \\ &= 68.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1436 [23.5 / 567] \\ &= 59.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(68.5, 59.5) \\ &= 59.469 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (63.6 + 950 + 447) / 56.3 \\ &= 25.942 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.05 \cdot 567 / 32 \\ &= 18.7 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 25.9 - 18.7, 18.7)
= 33.2 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	33.18	kgf/cm ²
Maximum Allowable External Pressure	Pmax	59.47	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -2 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : A
From Figure 3.7M- Non PWHT, the MDMT is : 21 °C
Temperature reduction per Graph 3.12 : 22 °C
The computed stress ratio is : 0.449

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (3.64 • 1)/(9.6 - 1.5 - 0)
= 0.449

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -23 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
Temperature reduction per Graph 3.12 : 22 °C
The computed stress ratio is : 0.449

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (3.64 • 1)/(9.6 - 1.5 - 0)
= 0.449

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -23 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
Temperature reduction per Graph 3.12 : 22 °C
The computed stress ratio is : 0.449

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (3.64 • 1)/(9.6 - 1.5 - 0)
= 0.449

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 4 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.12 : 10 °C
The computed stress ratio is : 0.765

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)



$$= (18 \cdot 1) / (25 - 1.5 - 0)$$

$$= 0.765$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -23 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
 Temperature reduction per Graph 3.12 : 22 °C
 The computed stress ratio is : 0.449

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.64 \cdot 1) / (9.6 - 1.5 - 0)$
 $= 0.449$

Gov. MDMT of the Nozzle : -2 °C
 Gov. MDMT of the Reinforcement Pad : 4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -45 °C

The Drop for this Nozzle is : 6.4737 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 331.4737 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	25.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1435.72	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm
Thickness of Reinforcing Pad	26.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	57.22	kgf/cm ²
Radial Load	P	-509.88	kgf
Circumferential Shear	Vc	764.82	kgf
Longitudinal Shear	Vl	764.82	kgf
Circumferential Moment	Mc	336.39	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	570.84	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{576.25(25.0 - 1.5)}$$

$$= 552.295 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1152.500 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	49.500 mm
Nozzle Thickness used	(t) =	9.473 mm
T / t	=	5.225
d / T	=	17.764

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.705$$

Nr/P	=	0.110
Mr/P	=	0.071
M0/P	=	0.181
N0/P	=	0.304
MrD/Mc	=	0.103
NrDL/Mc	=	0.126
M0d/Mc	=	0.297
N0DL/Mc	=	0.124
MrD/M1	=	0.102
NrDL/M1	=	0.122
M0D/M1	=	0.275
N0DL/M1	=	0.259

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-28	-28	28	28
Outplane Bending (ML)	-183	183	183	-183
Normal Pressure Stress	610	667	610	667
Outplane Stress Summary	426	806	851	496

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-67	67	67	-67
Inplane Pressure Stress	304	304	304	304
Inplane Stress Summary	234	352	397	244

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	25	25	25	25
Shear Stress Summary	31	31	20	20

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	431	808	852	497

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-145	145	145	-145
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	304	304	304	304
Outplane Stress Summary	178	423	489	153

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-50	50	50	-50
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	610	667	610	667
Inplane Stress Summary	560	700	681	620

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	25	25	25	25
Shear Stress Summary	20	20	31	31

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	561	702	687	622

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-28	-28	28	28



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	508	577	508	577
Hoop Stress Summary	486	555	543	612

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-258	-258	258	258
Axial Bending (ML)	-744	744	744	-744
Axial Pressure Stress	254	254	254	254
Axial Stress Summary	-682	696	1322	-275

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	30	30	-30	-30
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	135	135	135	135
Shear Stress Summary	165	165	104	104

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1214	805	1336	912

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	508	577	508	577
Hoop Stress Summary	505	573	525	593

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	-189	-189	189	189
Axial Bending (Mc)	-537	537	537	-537
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	254	254	254	254
Axial Stress Summary	-407	558	1047	-138

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-30	-30	30	30
Shear due to Torsion	135	135	135	135
Shear Stress Summary	104	104	165	165

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	935	671	1095	804

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(563.25(25.0 - 1.5))}$
 $= 547.938 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1126.500 mm
Nozzle Outside Diameter	(d) =	268.275 mm
Vessel Thickness used	(T) =	23.500 mm
Nozzle Thickness used	(t) =	59.473 mm
T / t	=	0.395
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.649$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.077
Mr/P	=	0.164
M0/P	=	0.048
N0/P	=	0.101
MrD/Mc	=	0.294
NrDL/Mc	=	0.094
M0d/Mc	=	0.091
N0DL/Mc	=	0.064
MrD/Ml	=	0.132
NrDL/Ml	=	0.081
M0D/Ml	=	0.042
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-37	-37	37	37
Outplane Bending (ML)	-78	78	78	-78
Normal Pressure Stress	1314	1371	1314	1371
Outplane Stress Summary	1234	1395	1466	1314

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-24	-24	24	24
Inplane Bending (ML)	-245	245	245	-245
Inplane Pressure Stress	657	657	657	657
Inplane Stress Summary	484	794	1025	353

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	29	29	13	13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1235	1396	1466	1314

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-124	124	124	-124
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	657	657	657	657
Outplane Stress Summary	554	749	831	530

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	-21	-21	21	21
Inplane Bending (Mc)	-400	400	400	-400
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1314	1371	1314	1371
Inplane Stress Summary	990	1667	1833	909

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	21	21	21	21
Shear Stress Summary	13	13	29	29

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 991 1667 1834 911

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	610	667	610	667	610	667	610	667
Circ. Pl (SUS)	-22	-22	34	34	-8	-8	12	12
Circ. Q (SUS)	-161	161	205	-205	-42	42	58	-58
Long. Pm (SUS)	304	304	304	304	304	304	304	304
Long. Pl (SUS)	-11	-11	15	15	-4	-4	16	16
Long. Q (SUS)	-59	59	75	-75	-123	123	167	-167
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	25	25	25	25	25	25	25	25
Pm (SUS)	610.0	667.0	610.0	667.0	610.0	667.0	610.0	667.0
Pm+Pl (SUS)	588.1	645.1	644.1	701.1	602.1	659.1	622.1	679.1
Pm+Pl+Q (Total)	431.6	808.0	849.9	497.6	561.0	702.4	684.6	622.9

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	667.00	1435.72	Passed
Pm+Pl (SUS)	701.07	2153.58	Passed
Pm+Pl+Q (TOTAL)	849.88	4307.17	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1314	1371	1314	1371	1314	1371	1314	1371
Circ. Pl (SUS)	-28	-28	46	46	-14	-14	28	28
Circ. Q (SUS)	-52	52	104	-104	-310	310	490	-490
Long. Pm (SUS)	657	657	657	657	657	657	657	657
Long. Pl (SUS)	-17	-17	31	31	-5	-5	23	23
Long. Q (SUS)	-155	155	335	-335	-98	98	150	-150
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	1314	1371	1314	1371	1314	1371	1314	1371
Pm+Pl (SUS)	1286	1343	1360	1417	1300	1357	1342	1399
Pm+Pl+Q (Total)	1235	1396	1464	1313	990	1667	1832	911

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	1371.00	1435.72	Passed
Pm+Pl (SUS)	1417.07	2153.58	Passed
Pm+Pl+Q (TOTAL)	1832.78	4307.17	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	508	577	508	577	508	577	508	577
Circ. Pl (SUS)	-22	-22	34	34	-4	-4	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	254	254	254	254	254	254	254	254
Long. Pl (SUS)	-248	-248	268	268	-179	-179	199	199
Long. Q (SUS)	-690	690	798	-798	-483	483	591	-591
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	30	30	-30	-30	-30	-30	30	30
Shear Q (SUS)	135	135	135	135	135	135	135	135
Pm (SUS)	508.0	577.0	508.0	577.0	508.0	577.0	508.0	577.0
Pm+Pl (SUS)	487.9	556.6	563.6	620.2	506.1	574.8	535.0	599.2
Pm+Pl+Q (Total)	1215	804	1333	911	935	670	1091	802

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	577.00	1203.20	Passed
Pm+Pl (SUS)	620.17	1804.80	Passed
Pm+Pl+Q (TOTAL)	1333.92	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	57.105	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1435.72	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	3142.4668	mm
Shell Finished (Minimum) Thickness	t	25.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		298.79	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1398.99	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

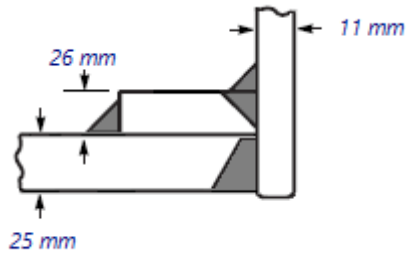
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		146.329	mm.
Actual Thickness Used in Calculation		10.973	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 149 \cdot (\exp(57.1/(1203 \cdot 1)) - 1) + 1.5 + 0$$

$$= 3.63 + 1.5 + 0 = 5.13 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1203 \cdot 1 \cdot \ln((2 \cdot 9.47 + 149) / 149) - 0$$

$$= 144 - 0 = 144 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.720 mm
Min. required shell thickness, internal p + c :	23.881 mm
Min. required shell thickness, external p + c :	8.393 mm

Nozzle neck minimum required thickness	: 7.720 mm
--	------------

Minimum nozzle neck thickness 9.601 mm , is greater than required thickness 7.720 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	615.2,	Allowable :	1203.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	2392.8 kgf/cm ²	Passed
Occasional :	249.6,	Allowable :	1600.3 kgf/cm ²	Passed
Shear :	258.1,	Allowable :	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1203/1436, 1]$$

$$= 0.838$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1399/1436, 1]$$

$$= 0.974$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 11 - 1.5$$

$$= 9.473 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1103/23.5$$

$$= 46.936 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1100/2 + 1.5 \\ &= 551.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned} \text{Limit [LR1]:} \\ &= \sqrt{(R_{\text{eff}} \cdot t) + W} \\ &= \sqrt{(552 \cdot 23.5) + 50} \\ &= 163.843 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR2]:} \\ &= \sqrt{((R_{\text{eff}} + t) \cdot (t + t_e))} \\ &= \sqrt{((552 + 23.5) \cdot (23.5 + 26))} \\ &= 168.708 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Limit [LR3]:} \\ &= 2 \cdot R_n \\ &= 2 \cdot 74.7 \\ &= 149.329 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Final value of [LR]:} \\ &= \min(LR1, LR2, LR3) \\ &= \min(164, 169, 149) \\ &= 149.329 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH1]:} \\ &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 23.5, 26] + \sqrt{(74.7 \cdot 9.47)} \\ &= 52.595 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH2]:} \\ &= L_{pr1} \\ &= 300 \\ &= 300.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Thickness Limit Candidate [LH3]:} \\ &= 8(t + t_e) \\ &= 8(23.5 + 26) \\ &= 396.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[52.6, 300, 396] + 23.5 \\ &= 76.095 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Effective Vessel Thickness [teff]:} \\ &= t + ((A5 \cdot frp) / LR) \\ &= 23.5 + ((13 \cdot 0.97) / 149) \\ &= 31.983 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Determine Parameter } [\lambda]: \\ &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{\text{eff}}) \cdot t_{\text{eff}}})) \\ &= \min(12, (2 \cdot 74.7 + 9.47) / (\sqrt{(1103 + 32) \cdot 32})) \\ &= 0.833 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned} \text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 23.5 \cdot 149 \cdot \max((0.83/5)^{0.85}, 1) \\ &= 35.092 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot LH \\ &= 9.47 \cdot 76.1 \\ &= 7.208 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 20^2 \\ &= 2.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e , LR \cdot t_e) \\ &= \min(50 \cdot 26, 149 \cdot 26) \\ &= 13.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 35.1 + 0.84(7.21 + 0) + 0.5 + 2 + 0 + 0.97(13) \\ &= 56.301 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 9.47 / \ln[1 + 9.47/74.7] \\ &= 79.307 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 32 / \ln[1 + 32/552] \\ &= 567.341 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1436 \cdot 1 \\ &= 2153.6 \text{ kgf/cm}^2 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 57.1 \cdot 79.3 \cdot 76.1 \\ &= 3446.2 \text{ kgf} \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 57.1 \cdot 567 (149 + 9.47) \\ &= 51449.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 57.1 \cdot 567 \cdot 74.7 \\ &= 24190.0 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (51449 + 24190 + 3446) / 57.1 \\ &= 1384.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2154 / (2 \cdot 1385 / 56.3 - 567 / 32) \\ &= 68.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S [t / R_{xs}] \\ &= 1436 [23.5 / 567] \\ &= 59.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(68.5, 59.5) \\ &= 59.469 \text{ kgf/cm}^2 \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (3446 + 51449 + 24190) / 56.3 \\ &= 1404.698 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 57.1 \cdot 567 / 32 \\ &= 1013.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1405 - 1013, 1013) \\ &= 1796.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	1796.41	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	59.47	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (74.7 + 9.47) / 74.7 \\ &= 1.127 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (74.7 + 9.47) \\ &= 132.163 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (74.7 + 9.47 + 50) \\ &= 210.703 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 14.142, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 9.473, 6.000) \text{ Passed} \\ &tf1, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 26.000, 0.6 \cdot 23.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(24190 \cdot 1.13, 1.5 \cdot 1203(7.21+0), \pi/4 \cdot 57.1 \cdot 74.7^2 \cdot 1.13^2) \\ &= 3175.017 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 3175 \cdot 23.5 \cdot 1436 / (23.5 \cdot 1436 + 26 \cdot 1399) \\ &= 1527.865 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 3175 \cdot 26 \cdot 1399 / (23.5 \cdot 1436 + 26 \cdot 1399) \\ &= 1647.151 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 1528 / (132 \cdot (0.6 \cdot 23.5 + 0.49 \cdot 0)) \\ &= 81.989 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 1647 / (132 \cdot (0.6 \cdot 26 + 0.49 \cdot 7.07)) \\ &= 65.372 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned}
 &= fwp / (L_t P \cdot (0.49 \cdot L_{42T})) \\
 &= 1647 / (211 \cdot (0.49 \cdot 14.1)) \\
 &= 112.812 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(82, 65.4, 113) \\
 &= 112.8 \text{ must be less than or equal to } 1435.7 \text{ kgf/cm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n / R_n] \\
 &= 9.47 / \ln[1 + 9.47 / 74.7] \\
 &= 79.307 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\
 &= 32 / \ln[1 + 32 / 552] \\
 &= 567.341 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1436 \\
 &= 2153.6 \text{ kgf/cm}^2
 \end{aligned}$$

Determine Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.05 \cdot 79.3 \cdot 76.1 \\
 &= 63.6 \text{ kgf}
 \end{aligned}$$

Determine Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 1.05 \cdot 567 (149 + 9.47) \\
 &= 950.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 1.05 \cdot 567 \cdot 74.7 \\
 &= 446.7 \text{ kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (950 + 447 + 63.6) / 1.05 \\
 &= 1384.9 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 2154 / (2 \cdot 1385 / 56.3 - 567 / 32) \\
 &= 68.5 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 1436 [23.5 / 567] \\
 &= 59.5 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(68.5, 59.5) \\
 &= 59.469 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (63.6 + 950 + 447) / 56.3 \\
 &= 25.942 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 1.05 \cdot 567 / 32 \\
 &= 18.7 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:
= max($2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}$, σ_{Circ})
= max($2 \cdot 25.9 - 18.7$, 18.7)
= 33.2 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	33.18	kgf/cm ²
Maximum Allowable External Pressure	Pmax	59.47	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -2 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A
From Figure 3.7M- Non PWHT, the MDMT is	: 21 °C
Temperature reduction per Graph 3.12	: 22 °C
The computed stress ratio is	: 0.448

The computed thickness ratio [sr]:
= ($tr \cdot E^*$) / ($tnom - ci - cext$)
= ($3.63 \cdot 1$) / ($9.6 - 1.5 - 0$)
= 0.448

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -23 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 22 °C
The computed stress ratio is	: 0.448

The computed thickness ratio [sr]:
= ($tr \cdot E^*$) / ($tnom - ci - cext$)
= ($3.63 \cdot 1$) / ($9.6 - 1.5 - 0$)
= 0.448

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -23 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 22 °C
The computed stress ratio is	: 0.448

The computed thickness ratio [sr]:
= ($tr \cdot E^*$) / ($tnom - ci - cext$)
= ($3.63 \cdot 1$) / ($9.6 - 1.5 - 0$)
= 0.448

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: 4 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: 14 °C
Temperature reduction per Graph 3.12	: 10 °C
The computed stress ratio is	: 0.763

The computed thickness ratio [sr]:
= ($tr \cdot E^*$) / ($tnom - ci - cext$)



$$= (17.9 \cdot 1) / (25 - 1.5 - 0)$$

$$= 0.763$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -23 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
 Temperature reduction per Graph 3.12 : 22 °C
 The computed stress ratio is : 0.448

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.63 \cdot 1) / (9.6 - 1.5 - 0)$
 $= 0.448$

Gov. MDMT of the Nozzle : -2 °C
 Gov. MDMT of the Reinforcement Pad : 4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -45 °C

The Drop for this Nozzle is : 6.4737 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 331.4737 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	25.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1435.72	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm
Thickness of Reinforcing Pad	26.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	57.11	kgf/cm ²
Radial Load	P	-509.88	kgf
Circumferential Shear	Vc	764.82	kgf
Longitudinal Shear	Vl	764.82	kgf
Circumferential Moment	Mc	336.39	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	570.84	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{576.25(25.0 - 1.5)}$$

$$= 552.295 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1152.500 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	49.500 mm
Nozzle Thickness used	(t) =	9.473 mm
T / t	=	5.225
d / T	=	17.764

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.705$$

Nr/P	=	0.110
Mr/P	=	0.071
M0/P	=	0.181
N0/P	=	0.304
MrD/Mc	=	0.103
NrDL/Mc	=	0.126
M0d/Mc	=	0.297
N0DL/Mc	=	0.124
MrD/M1	=	0.102
NrDL/M1	=	0.122
M0D/M1	=	0.275
N0DL/M1	=	0.259

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-28	-28	28	28
Outplane Bending (ML)	-183	183	183	-183
Normal Pressure Stress	608	665	608	665
Outplane Stress Summary	425	804	850	494

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-67	67	67	-67
Inplane Pressure Stress	304	304	304	304
Inplane Stress Summary	234	352	396	243

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	25	25	25	25
Shear Stress Summary	31	31	20	20

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	430	806	851	496

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-145	145	145	-145
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	304	304	304	304
Outplane Stress Summary	178	423	488	153

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-50	50	50	-50
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	608	665	608	665
Inplane Stress Summary	559	699	680	619

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	25	25	25	25
Shear Stress Summary	20	20	31	31

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	560	701	685	621

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-28	-28	28	28



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	507	576	507	576
Hoop Stress Summary	485	553	542	611

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-258	-258	258	258
Axial Bending (ML)	-744	744	744	-744
Axial Pressure Stress	253	253	253	253
Axial Stress Summary	-682	695	1321	-276

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	30	30	-30	-30
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	135	135	135	135
Shear Stress Summary	165	165	104	104

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1214	805	1335	912

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	507	576	507	576
Hoop Stress Summary	504	572	524	592

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	-189	-189	189	189
Axial Bending (Mc)	-537	537	537	-537
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	253	253	253	253
Axial Stress Summary	-407	558	1046	-138

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-30	-30	30	30
Shear due to Torsion	135	135	135	135
Shear Stress Summary	104	104	165	165

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	935	670	1095	803

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(563.25(25.0 - 1.5))}$
 $= 547.938 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1126.500 mm
Nozzle Outside Diameter	(d) =	268.275 mm
Vessel Thickness used	(T) =	23.500 mm
Nozzle Thickness used	(t) =	59.473 mm
T / t	=	0.395
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.649$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.077
Mr/P	=	0.164
M0/P	=	0.048
N0/P	=	0.101
MrD/Mc	=	0.294
NrDL/Mc	=	0.094
M0d/Mc	=	0.091
N0DL/Mc	=	0.064
MrD/Ml	=	0.132
NrDL/Ml	=	0.081
M0D/Ml	=	0.042
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-37	-37	37	37
Outplane Bending (ML)	-78	78	78	-78
Normal Pressure Stress	1312	1369	1312	1369
Outplane Stress Summary	1231	1392	1463	1311

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-24	-24	24	24
Inplane Bending (ML)	-245	245	245	-245
Inplane Pressure Stress	656	656	656	656
Inplane Stress Summary	483	793	1024	351

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	29	29	13	13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1232	1394	1464	1311

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-124	124	124	-124
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	656	656	656	656
Outplane Stress Summary	552	748	830	529

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	-21	-21	21	21
Inplane Bending (Mc)	-400	400	400	-400
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1312	1369	1312	1369
Inplane Stress Summary	988	1664	1831	906

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	21	21	21	21
Shear Stress Summary	13	13	29	29

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 988 1664 1832 909

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	608	665	608	665	608	665	608	665
Circ. Pl (SUS)	-22	-22	34	34	-8	-8	12	12
Circ. Q (SUS)	-161	161	205	-205	-42	42	58	-58
Long. Pm (SUS)	304	304	304	304	304	304	304	304
Long. Pl (SUS)	-11	-11	15	15	-4	-4	16	16
Long. Q (SUS)	-59	59	75	-75	-123	123	167	-167
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	25	25	25	25	25	25	25	25
Pm (SUS)	608.0	665.0	608.0	665.0	608.0	665.0	608.0	665.0
Pm+Pl (SUS)	586.1	643.1	642.1	699.1	600.1	657.1	620.1	677.1
Pm+Pl+Q (Total)	429.6	806.0	847.9	495.6	559.0	700.4	682.6	620.9

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	665.00	1435.72	Passed
Pm+Pl (SUS)	699.07	2153.58	Passed
Pm+Pl+Q (TOTAL)	847.88	4307.17	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1312	1369	1312	1369	1312	1369	1312	1369
Circ. Pl (SUS)	-28	-28	46	46	-14	-14	28	28
Circ. Q (SUS)	-52	52	104	-104	-310	310	490	-490
Long. Pm (SUS)	656	656	656	656	656	656	656	656
Long. Pl (SUS)	-17	-17	31	31	-5	-5	23	23
Long. Q (SUS)	-155	155	335	-335	-98	98	150	-150
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	1312	1369	1312	1369	1312	1369	1312	1369
Pm+Pl (SUS)	1284	1341	1358	1415	1298	1355	1340	1397
Pm+Pl+Q (Total)	1233	1394	1462	1311	988	1665	1830	909

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	1369.00	1435.72	Passed
Pm+Pl (SUS)	1415.07	2153.58	Passed
Pm+Pl+Q (TOTAL)	1830.78	4307.17	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	507	576	507	576	507	576	507	576
Circ. Pl (SUS)	-22	-22	34	34	-4	-4	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	253	253	253	253	253	253	253	253
Long. Pl (SUS)	-248	-248	268	268	-179	-179	199	199
Long. Q (SUS)	-690	690	798	-798	-483	483	591	-591
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	30	30	-30	-30	-30	-30	30	30
Shear Q (SUS)	135	135	135	135	135	135	135	135
Pm (SUS)	507.0	576.0	507.0	576.0	507.0	576.0	507.0	576.0
Pm+Pl (SUS)	486.9	555.6	562.6	619.2	505.1	573.8	534.0	598.2
Pm+Pl+Q (Total)	1215	803	1332	911	935	669	1090	802

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	576.00	1203.20	Passed
Pm+Pl (SUS)	619.17	1804.80	Passed
Pm+Pl+Q (TOTAL)	1332.92	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1-Vent

From: 50

Pressure for Reinforcement Calculations	P	57.105	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1435.72	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	3142.4668	mm
Shell Finished (Minimum) Thickness	t	25.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		518.79	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0744	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

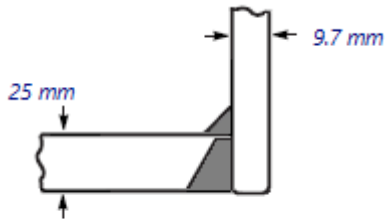
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1-Vent

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		40.945	mm.
Actual Thickness Used in Calculation		9.690	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 43.9 \cdot (\exp(57.1/(1203 \cdot 1)) - 1) + 1.5 + 0$
 $= 1.07 + 1.5 + 0 = 2.57 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1203 \cdot 1 \cdot \ln((2 \cdot 8.19 + 43.9) / 43.9) - 0$
 $= 381 - 0 = 381 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	4.920 mm
Min. required shell thickness, internal p + c :	23.881 mm
Min. required shell thickness, external p + c :	8.393 mm

Nozzle neck minimum required thickness :	4.920 mm
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Minimum nozzle neck thickness 9.703 mm , is greater than required thickness 4.920 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1203/1436, 1]$
 $= 0.838$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 9.69 - 1.5$
 $= 8.190 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1103/23.5$
 $= 46.936 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1100/2 + 1.5$
 $= 551.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{552 \cdot 23.5}, 2 \cdot 22)$
 $= 43.945 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 23.5, 0] + \sqrt{22 \cdot 8.19}$
 $= 13.415 \text{ mm}$

Thickness Limit Candidate [LH2]:
 $= L_{pr1}$



$$= 300$$
$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(23.5 + 0)$$
$$= 188.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$
$$= \min[13.4, 300, 188] + 23.5$$
$$= 36.915 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 23.5 + ((0 \cdot 1) / 43.9)$$
$$= 23.500 \text{ mm}$$

Determine Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 22 + 8.19) / (\sqrt{(1103 + 23.5) \cdot 23.5}))$$
$$= 0.320$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 23.5 \cdot 43.9 \cdot \max((0.32/5)^{0.85}, 1)$$
$$= 10.327 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$
$$= 8.19 \cdot 36.9$$
$$= 3.023 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$
$$= 0.5 \cdot 16^2 - 0.033$$
$$= 1.247 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$
$$= 10.3 + 0.84(3.02 + 0) + 1.25 + 0 + 0$$
$$= 14.107 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 8.19 / \ln[1 + 8.19/22]$$
$$= 25.852 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 23.5 / \ln[1 + 23.5/552]$$
$$= 563.167 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1436 \cdot 1$$
$$= 2153.6 \text{ kgf/cm}^2$$

Determine Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 57.1 \cdot 25.9 \cdot 36.9$$
$$= 545.0 \text{ kgf}$$

Determine Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$
$$= 57.1 \cdot 563 (43.9 + 8.19)$$
$$= 16766.5 \text{ kgf}$$



Discontinuity Force from Internal Pressure [FY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 57.1 \cdot 563 \cdot 22 \\ &= 7066.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (16767 + 7066 + 545) / 57.1 \\ &= 426.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 2154 / (2 \cdot 427 / 14.1 - 563 / 23.5) \\ &= 58.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1436 [23.5/563] \\ &= 59.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(58.9, 59.9) \\ &= 58.912 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (545 + 16767 + 7066) / 14.1 \\ &= 1728.026 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 57.1 \cdot 563 / 23.5 \\ &= 1368.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1728 - 1369, 1369) \\ &= 2087.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	2087.54	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	58.91	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (22 + 8.19) / 22 \\ &= 1.373 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (22 + 8.19) \\ &= 47.379 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 11.314, \text{ must be } \geq \min(0.7 \cdot 8.190, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(7066 \cdot 1.37, 1.5 \cdot 1203(3.02 + 0), \pi / 4 \cdot 57.1 \cdot 22^2 \cdot 1.37^2) \\ &= 408.039 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= $408 / (47.4 (0.49 \cdot 11.3 + 0.6 \cdot 11.1 + 0.49 \cdot 0))$
= 70.660 < or = to 1435.722 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $8.19 / \ln[1 + 8.19/22]$
= 25.852 mm

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $23.5 / \ln[1 + 23.5/552]$
= 563.167 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 1436$
= 2153.6 kgf/cm²

Determine Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.05 \cdot 25.9 \cdot 36.9$
= 10.1 kgf

Determine Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.05 \cdot 563 (43.9 + 8.19)$
= 309.6 kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.05 \cdot 563 \cdot 22$
= 130.5 kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
= $(310 + 130 + 10.1) / 1.05$
= 426.9 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $2154 / (2 \cdot 427 / 14.1 - 563 / 23.5)$
= 58.9 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
= $1436 [23.5 / 563]$
= 59.9 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(58.9, 59.9)$
= 58.912 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:

= $(f_N + f_S + f_Y) / A_T$
= $(10.1 + 310 + 130) / 14.1$
= 31.913 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$
= $1.05 \cdot 563 / 23.5$
= 25.3 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 31.9 - 25.3, 25.3)$
= 38.6 kgf/cm²



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2153.58	kgf/cm ²
Local Primary Membrane Stress	PL	38.55	kgf/cm ²
Maximum Allowable External Pressure	Pmax	58.91	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.130

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (1.07 \cdot 1) / (9.69 - 1.5 - 0) \\ &= 0.130 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -45 °C

The Drop for this Nozzle is : 0.8277 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.8277 mm

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	57.11	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	57.11	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	25.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1100.000	mm
Shell Temperature for Internal Pressure	Ts	270.00	°C
Shell Material		SA-516 70	
Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1435.80	kgf/cm ²
Shell Allowable Stress at Ambient		1784.51	kgf/cm ²

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	66.0000	mm
Channel Internal Corrosion Allowance	cac	1.5000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1100.000	mm
Channel Design Temperature	TEMPC	300.00	°C
Channel Flange Matl		SA-387 11	
Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	1672.36	kgf/cm ²
Channel Allowable Stress at Ambient		2110.82	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1452	
Tube Wall Thickness	et	2.5000	mm
Tube Outside Diameter	D	19.0000	mm
Total Straight Tube Length	Lt	2600.00	mm
Straight Tube Length (bet. inner tubsht faces) L		2370.00	mm
Design Temperature of the Tubes		300.00	°C
Tube Material		SA-213 T11	
Tube Material UNS Number		K11597	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1111.45	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1672.28	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.5000	mm
Groove Weld Leg	ag	2.5000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	531.045	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	300.00	°C
Tubesheet Material		SA-336 F11	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	230.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1430.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 1.0546 kgf/cm ²

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Shell
Flange Outside Diameter	A 1430.000 mm
Flange Inside Diameter	B 1100.000 mm
Flange Face Outside Diameter	Fod 1175.000 mm
Flange Face Inside Diameter	Fid 1100.000 mm
Gasket Outside Diameter	Go 1164.000 mm
Gasket Inside Diameter	Gi 1132.000 mm
Small end Hub thk.	g0 25.0000 mm
Large end Hub thk.	g1 40.0000 mm
Gasket Factor,	m 0.00
Gasket Design Seating Stress	y 0.00 kgf/cm ²
Flange Facing Sketch	Code Sketch 1a
Column for Gasket Seating	Code Column II
Gasket Thickness	tg 3.1750 mm
Flange Face Nubbin Width	w 15.0000 mm
Full face Gasket Flange Option	Program Selects

Bolting Information:

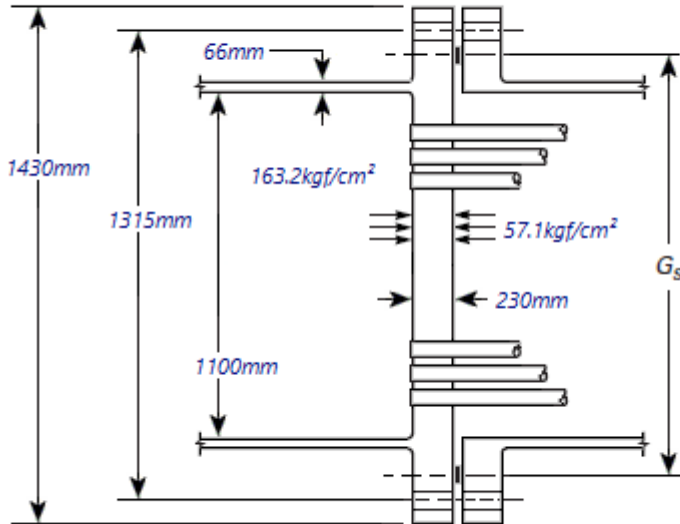
Diameter of Bolt Circle	C 1315.000 mm
Nominal Bolt Diameter	dB 48.0000 mm
Thread Series	TEMA Metric Thread
Number of Bolts	n 36
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 1753.82 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa 1753.82 kgf/cm ²
Weld between Flange and Shell/Channel	0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 214.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes



Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME Division 2 Tubesheet Results per Part 4.18, 2023

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	270.0 °C	0.19093E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	300.0 °C	0.19011E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm²

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	300.0 °C	0.19011E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm²

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	300.0 °C	0.19011E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(164/(1111 \cdot 1)) - 1) + 0$
 $= 1.11 + 0$
 $= 1.11 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 2901 / \sqrt{(9.5 \cdot 2.5)}$
 $= 595.282$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$



$$= 0.55 \cdot (19/2.5)$$

$$= 0.0724$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t/Do)$$

$$= 1.6 \cdot 0.072 \cdot 1901117(2.5/19)$$

$$= 28964.391 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 28964.4/1672.3$$

$$= 17.320$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 1672/4218$$

$$= 0.396$$

Ratio [AE]:

$$= Fhe / E$$

$$= 28964/1901117$$

$$= 0.015235$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.4) + (0.20000E-04 - 0.002)) / (\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.201$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 1672(1 + 0.0020) / \ln(1 + 0.0020) 0.20139404$$

$$= 5858.949 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1 - 0.4)$$

$$= 0.362$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (4218 \cdot \exp(0.36)) / (0.36 0.36)$$

$$= 8753.396 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.41.5 - 0.5 \cdot 0.42.5 - 0.43.5$$

$$= 0.286$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-0.799	0.737E-05	0.681E-07	0.129E-04	0.660E-05	0.365E+05	0.426E-01
2	1054	-3.698	0.472E-06	0.471E-06	0.379E-05	-0.378E-05	0.674E+06	0.137E-01
3	1581	-2.248	0.236E-05	0.221E-05	0.775E-05	-0.705E-05	0.172E+06	0.606E-02
4	1845	-1.524	0.434E-05	0.318E-05	0.102E-04	-0.601E-05	0.819E+05	0.729E-02
5	1713	-1.886	0.324E-05	0.282E-05	0.892E-05	-0.718E-05	0.120E+06	0.971E-03
6	1779	-1.705	0.376E-05	0.306E-05	0.954E-05	-0.683E-05	0.995E+05	0.265E-02
7	1746	-1.795	0.349E-05	0.295E-05	0.923E-05	-0.706E-05	0.109E+06	0.727E-03
8	1730	-1.841	0.336E-05	0.289E-05	0.907E-05	-0.713E-05	0.115E+06	0.148E-03
9	1738	-1.818	0.343E-05	0.292E-05	0.915E-05	-0.710E-05	0.112E+06	0.282E-03
10	1734	-1.829	0.339E-05	0.290E-05	0.911E-05	-0.711E-05	0.113E+06	0.654E-04
11	1732	-1.835	0.338E-05	0.290E-05	0.909E-05	-0.712E-05	0.114E+06	0.418E-04
12	1733	-1.832	0.339E-05	0.290E-05	0.910E-05	-0.712E-05	0.114E+06	0.117E-04
13	1732	-1.834	0.338E-05	0.290E-05	0.910E-05	-0.712E-05	0.114E+06	0.151E-04
14	1733	-1.833	0.338E-05	0.290E-05	0.910E-05	-0.712E-05	0.114E+06	0.172E-05
15	1733	-1.833	0.338E-05	0.290E-05	0.910E-05	-0.712E-05	0.114E+06	0.498E-05



16	1733	-1.833	0.338E-05	0.290E-05	0.910E-05	-0.712E-05	0.114E+06	0.163E-05
----	------	--------	-----------	-----------	-----------	------------	-----------	-----------

Factor of Safety [FS]:
Since $Fic [1733] \geq Fy[1672]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:
= Fic/FS
= $1733/1.67$
= $1039.657 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
= $2 \cdot Fha(t / Do)$
= $2 \cdot 1040 (2.5/19)$
= 273.594 kgf/cm^2

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1145 mm
Allowable Internal Pressure at Corroded thickness	339.42 kgf/cm ²
Required Internal Design Pressure	164.21 kgf/cm ²
Allowable External Pressure at Corroded thickness	273.59 kgf/cm ²
Required External Design Pressure	57.11 kgf/cm ²
Required Thickness due to Shell Side pressure	0.9136 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 2, 2023

Gasket Contact Width,	$N = (Goc - Gic) / 2$	16.000 mm
Basic Gasket Width,	$b0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = Go \text{ (Self-Energizing)}$	1164.000 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2023, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Compute the equivalent outer tube limit circle diameter [Do]:
= $2 \cdot ro + dt$
= $2 \cdot 531 + 19$
= 1081.090 mm

Determine the basic ligament efficiency for shear [μ]:
= $(p - dt) / p$
= $(25 - 19) / 25$
= 0.240

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Compute the ratio [ρ_s]:
= Gs / Do (Configurations d, e, f)
= $1164 / 1081$
= 1.0767

Compute the ratio [ρ_c]:
= Dc / Do (Configurations a, e, f)
= $1100 / 1081$
= 1.0175

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Compute the ratio [ρ]:
= ltx / h
= 0/230



$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Compute the effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19 - 2 \cdot 2.5 (1901117/1901117) \cdot (1111/1407) \cdot (0); 19 - 2 \cdot 2.5)$$

$$= 19.0000 \text{ mm}$$

Compute the effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 25 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 1081 \cdot 25) / (\pi \cdot 1081^2)) }$$

$$= 25.0000 \text{ mm}$$

Compute the effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25 - 19) / 25$$

$$= 0.2400$$

$$h/p = 9.200001 ; \mu^* = 0.240000 ; E^* = 387883 \text{ kgf/cm}^2$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$

Note: As h/p (9.200) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s and ω_s are all = 0

Determine channel coefficient [β_c]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (Dc + tc) tc }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1100 + 66) 66 }$$

$$= 0.0066 \text{ 1/mm}$$

Determine channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6 (1 - \nu_s^2))$$

$$= 0.0066 \cdot 1901117 \cdot 66^3 / (6 (1 - 0.3^2))$$

$$= 655967680.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Determine channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CC = CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Determine channel coefficient [λ_c]:

$$= 6 (Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6 (1100 + 66 / \cos(0)) \cdot 655967680 / 230^3 \cdot (1 + 230 \cdot 0.0066 \cdot (1 + 1) / 2 + (230 \cdot 0.0066)^2 / 2 \cdot 1)$$

$$= 1374048.2 \text{ kgf/cm}^2$$

Determine channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot E_c \cdot tc) \cdot (1 - Dc \cdot \nu_c / (2 (Dc + tc)))$$

$$= 1100^2 / (4 \cdot 1901117 \cdot 66) \cdot (1 - 1100 \cdot 0.3 / (2 (1100 + 66)))$$

$$= 0.20697E-02 \text{ mm/kgf/cm}^2$$

Determine channel coefficient [zetac]:

$$\text{zetac} = 0$$

Calculate parameter [ω_c]:

$$= \rho_c \cdot kc \cdot \beta_c \cdot (\delta_c (CmTheta + CqDelta \cdot h \cdot \beta_c) - \zeta_c \cdot (CmTheta / \beta_c + h \cdot CqTheta / 2))$$

$$= 1.02 \cdot 655967680 \cdot 0.0066 \cdot (0.002069703 (1 + 1 \cdot 230 \cdot 0.0066) - 0.000 \cdot (1 / 0.0066 + 230 \cdot 1/2))$$

$$= 22695.4883 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 1430 / 1081$$

$$= 1.3227$$



Compute coefficient [F]:

$$\begin{aligned} &= (1 - v) / E \cdot (\lambda c + E \cdot \ln(K)) \\ &= (1 - 0.41) / 387884 \cdot (1374048 + 1901117 \cdot \ln(1.32)) \\ &= 2.9127 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(\rho s - 1) \cdot (\rho s^2 + 1) \cdot Ps - (\rho c - 1) \cdot (\rho c^2 + 1) \cdot Pt] \\ &= 1081^2/16 \cdot [(1.08 - 1) \cdot (1.08^2 + 1) \cdot 57.1 - \\ &\quad (1.02 - 1) \cdot (1.02^2 + 1) \cdot 163] \\ &= 266471.9062 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + (\omega c \cdot Pt) + W \cdot (C - Gs) / (2 \cdot \pi \cdot Do) \\ &= 266472 + (3702950) + 847910 \cdot (1315 - 1164) / (2 \cdot \pi \cdot 1081) \\ &= 5854311.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M) - Do^2/32 \cdot F(Ps - Pt) / (1 + F) \\ &= (5854311) - 1081^2/32 \cdot 2.91(57.1 - 163) / (1 + 2.91) \\ &= 4379694.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + v) \cdot (Ps - Pt) \\ &= 4379694 + 1081^2/64 \cdot (3 + 0.41) \cdot (57.1 - 163) \\ &= -2219037.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|4379694|; |-2219037|) \\ &= 4379694.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\ &= 6 \cdot 4379694 / (0.24(230 - 6)^2) \\ &= 2182.1655 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1407 \\ &= 2814.26 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\ &= 6 \cdot 4242946 / (0.24(200 - 6)^2) \\ &= 2809.6641 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 200 + 0 + 0 \\ &= 200.3017 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(200; 106) \\ &= 200.3017 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned} &= |Ps - Pt| \\ &= |57.1 - 163| \\ &= 106.053 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [Tlimit]:



$$\begin{aligned} &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\ &= 4 \cdot 0.24 \cdot 230 / 1081 \cdot \min(0.8 \cdot 1407; 0.533 \cdot 2233) \\ &= 229.912 \text{ kgf/cm}^2 \end{aligned}$$

$|Ps - Pt|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (Do / \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc \\ &= 1 / (4 \cdot 0.24) \cdot (1081 / 1126) \cdot |57.1 - 163| + 0 \\ &= 106.0933 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 66.000 mm
Min. Channel len. adjacent to the tubesheet is: 484.999mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$\begin{aligned} &= Dc^2 / (4 \cdot tc(Dc + tc) \cdot \cos(\theta)) \cdot Pt \\ &= 1100^2 / (4 \cdot 66(1100 + 66) \cdot \cos(0)) \cdot 163 \\ &= 641.344 \text{ kgf/cm}^2 \end{aligned}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$\begin{aligned} &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - \\ &\quad 6((1 - \nu) / E) \cdot Do / h^3 \cdot (Cm\theta + \\ &\quad h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\ &= 6 \cdot 14461603 / 66^2 \cdot [(0.002069703 \cdot 1 \cdot 0.0066 - 0.000 \cdot 1) \cdot 163 - \\ &\quad 6((1 - 0.41) / 387884) \cdot 1081 / 230^3 \cdot (1 + \\ &\quad 0.0066 \cdot 230 / 2 \cdot 1) \cdot (4379694 + 1081^2 / 32 \cdot (57.1 - 163))] \\ &= 1345.798 \text{ kgf/cm}^2 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned} &= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot Sc) \\ &= 1987.143 \text{ kgf/cm}^2 \text{ (Should be } \leq 2508.54) \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (19 - 2.5) 2.5 \\ &= 1.296 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) Sa \\ &= \pi \cdot 2.5 (19 - 2.5) 1111 \\ &= 1440.337 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft) \\ &= 0.85 \cdot \pi \cdot 2.5 (19 + 0.67 \cdot 2.5) 1111 \\ &= 1440.3367 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= Sa / Sw \text{ where } Sw = \min(Sa \text{ and } St) \\ &= 1111 / 1111 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft) \\ &= 0.55 \cdot \pi \cdot 2.5 (19 + 0.67 \cdot 2.5) 1111 \\ &= 992.6321 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} \right) - 0.75 \cdot do \\ &= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2.5 \cdot (19 - 2.5) \cdot 1 \cdot 1} \right) - \\ &\quad 0.75 \cdot 19 \\ &= 2.945 \text{ mm} \end{aligned}$$



Minimum Required Fillet Weld Leg [afr]: 1.5000 mm
 Minimum Required Groove Weld Leg [agr]: 1.5000 mm

Tube-Tubesheet Jt allowable, 1440 is >= tube strength 1440 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2182.2 <=	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	519.3 <=	1125.7 kgf/cm ²	Ok
Stress in Channel at Tubesheet	1987.1 <=	2508.5 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	200.302	230.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.500	2.500	Ok
Tube-Tubesheet Groove Weld Leg :	1.500	2.500	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2023

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	223.109	86.708	2608 2814 804 1126	Ps+Pt D1	Ok
D2uc	142.077	...	1325 2814 280 1126	Ps+Pt D2	Ok
D3uc	200.302	...	2182 2814 519 1126	Ps+Pt D3	Ok
D4uc	25.419	...	253 2814 5 1126	Ps+Pt D4	Ok
D1c	223.758	86.708	2616 2814 815 1126	Ps+Pt-c D1	Ok
D2c	144.112	...	1349 2814 283 1126	Ps+Pt-c D2	Ok
D3c	201.327	...	2197 2814 526 1126	Ps+Pt-c D3	Ok
D4c	26.834	...	259 2814 5 1126	Ps+Pt-c D4	Ok
Max:	223.7582	86.708 mm	0.930	0.724 (Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	3542	5017	Ok
D2uc	...	...	...	...	1538	2509	Ok
D3uc	...	...	...	...	1987	2509	Ok
D4uc	...	...	...	...	605	2509	Ok
D1c	...	...	...	...	3694	5017	Ok
D2c	...	...	...	...	1596	2509	Ok



D3c	...	...	...	...	2081	2509	Ok
D4c	...	...	...	...	628	2509	Ok
Max	...	...	...	...	0.829		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	223.758	230.000	Ok
Tubesheet thickness flanged extension :	86.708	214.000	Ok
Tube thickness :	1.115	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.500	Ok
Tube-tubesheet groove weld leg :	1.500	2.500	Ok

Given Shell Thickness: 25.0000 mm
 Given Channel Thickness: 66.0000 mm

Min Channel length of thk, (66.000) adj. to tubesheet: 484.999 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	173.781	1.000	122.424	0.995
Tubesheet Shear Stress	226.912	1.000	226.912	1.000
Tubesheet-Channel Junction	173.781	1.000	122.424	0.995
Tube Pressure Stress	339.416	1.000	273.593	1.000
Tubesheet Extension Stress	No Calc	No Calc	79.680	...
Minimum MAWP	173.781		79.680	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	174.736	1.000	127.877	1.000
Tubesheet Shear Stress	229.911	1.000	229.911	1.000
Tubesheet-Channel Junction	229.911	1.316	229.911	1.909
Tube Pressure Stress	367.435	1.000	326.845	1.000
Tubesheet Extension Stress	No Calc	No Calc	79.680	...
Minimum MAPnc	174.736		79.680	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:
 = max(tubesheet thk/4, min(tubesheet thk, channel thickness))
 = max(230/4, min(230, 66))
 = 66.000 mm

Note: This Element/Detail was specified as being Impact Tested.

Shell Side MDMT calculation:

Governing thickness for the shell side:
 = tubesheet thickness/4
 = 230/4
 = 57.500 mm

Note: This Element/Detail was specified as being Impact Tested.



where the MDMT reduction ratio is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(163/175, 57.1/79.7)
= 0.934

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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